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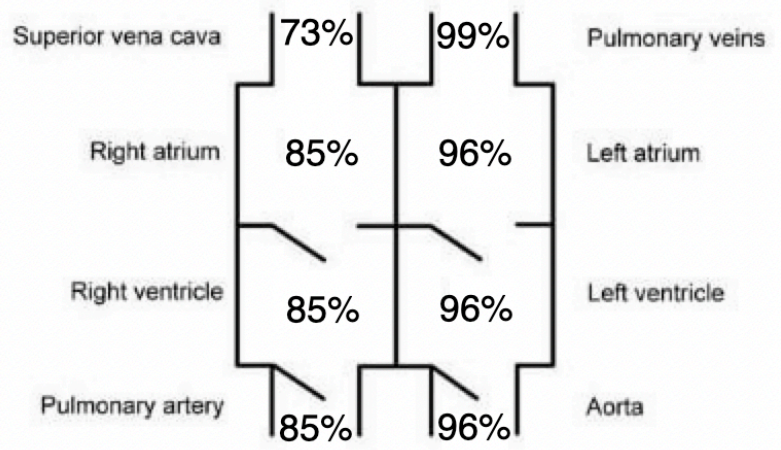
HY Cardio – by Dr Mike Mehlman

Important basic heart failure points	
Left heart failure	- Presents as pulmonary findings (i.e., pulmonary edema, dyspnea, orthopnea, paroxysmal nocturnal dyspnea). - This is because left-heart problems cause a backup of pressure onto the pulmonary circulation, leading to increased pulmonary capillary hydrostatic pressure → transudation of fluid into the alveolar spaces (pulmonary edema). Sometimes this can also cause pleural effusion. - Left atrial pressure (LAP) = pulmonary capillary wedge pressure (PCWP). - Therefore, if there is left heart pathology, PCWP is high (exceedingly HY). - Conversely, if a Q gives you normal PCWP, you know there's nothing wrong with the left heart. Qs will often give high PCWP and low BP, where you need to know immediately that means cardiogenic shock. - What USMLE will do is give you some sort of left-heart pathology + dyspnea, and then ask for the cause of the dyspnea → answer = "increased pulmonary capillary hydrostatic pressure." Another answer in this case is "increased alveolar-arteriolar (A-a) oxygen gradient."
Right heart failure	- Presents as systemic findings – i.e., jugular venous distension (JVD) and peripheral edema. - Since blood cannot enter the right heart as easily, it backs up to the neck veins (JVD) and venous circulation (increased hydrostatic pressure in veins → transudation of fluid into legs). The Q might mention that central venous pressure is high. - Hepatosplenomegaly can also be seen in RHF but is very rare on USMLE. - Normal jugular venous pressure (JVP) is 3cm above the sternal angle. JVD would be higher than this. Sometimes questions can write that jugular venous pulsations are seen 3cm above the sternal angle and the student erroneously thinks this means JVD, but this is not the case.
Congestive heart failure	- Congestive heart failure = left heart failure + right heart failure. - The most common cause of right heart failure is left heart failure. Simply adding the two together, we now call that congestive heart failure. - In congestive heart failure, we'll see both left- and right-heart failure findings – i.e., patient will have dyspnea, JVD, and peripheral edema. - PCWP is elevated in these patients, since the left heart has pathology.
Cor pulmonale	- Cor pulmonale is defined as right-heart failure due to a pulmonary cause. In other words, the left heart is completely normal in cor pulmonale and PCWP is normal. - Cor pulmonale will be a patient who has JVD and peripheral edema in the setting of obvious and overt lung disease, such as 100-pack-year smoking history, cystic fibrosis, or pulmonary fibrosis. These can present with lung findings such as wheezes, where you as the student need to say, "It just doesn't seem like they're focusing on left-heart failure as the cause of the right-heart failure here. It seems the 100-pack-year smoking Hx causing COPD is why the right heart is failing." - Can occur due to bronchopulmonary dysplasia in premature neonates or obstructive sleep apnea, since the lung disease backs up to the right heart. - The patient can have a "boot-shaped" heart colloquially, which refers to right ventricular hypertrophy without left ventricular hypertrophy. - If the patient has COPD, the massively hyperinflated lungs will push the heart to the midline, causing a long, narrow cardiac silhouette, with a point of maximal impulse in the sub-xiphoid space. - You must know that *pulmonary hypertension* is the reason the right heart decompensates. In both cor pulmonale and congestive heart failure, the

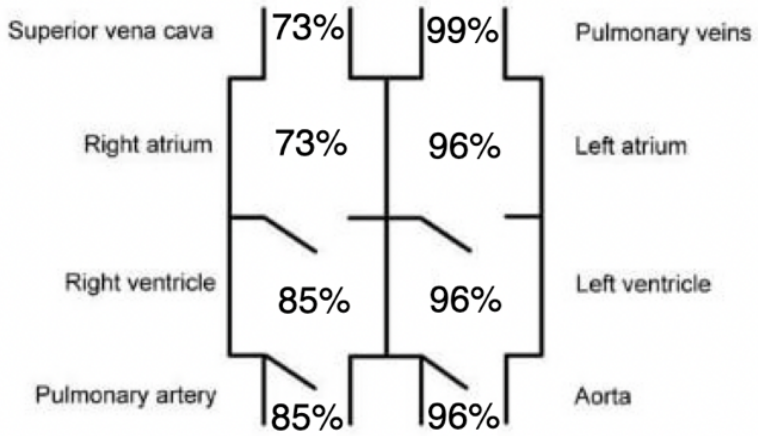
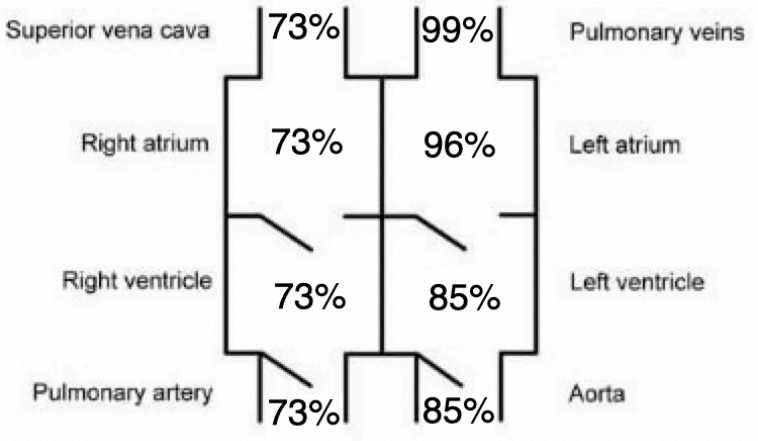
	right heart experiences increased afterload because of *pulmonary hypertension*. - Endothelin 1 is vasoconstrictor and key mediator in pulmonary hypertension. USMLE wants you to know this is increased in both cor pulmonale and left heart failure. Bosentin is an endothelin 1 receptor antagonist. - Nitric oxide synthase, in contrast, USMLE wants you to know is decreased in pulmonary hypertension (makes sense, since NO dilates). - A loud P2 and tricuspid regurgitation are HY findings in cor pulmonale. I discuss these in more detail in the tables below.

Hyper-quick causes of bilateral pitting peripheral edema	
Cardiac	- Right heart failure (either due to cor pulmonale or congestive) → ↓ ability to fill right heart → ↑ central venous pressure → ↑ **systemic venous hydrostatic pressure** → transudation of fluid from systemic veins/venules into interstitium of legs. - Tangential, but Step 1 NBME asks why HTN doesn't automatically cause peripheral edema → answer = "increased pre-capillary resistance." The USMLE wants you to know that arterioles are responsible for the majority of peripheral resistance; in the setting of high BP, the reason capillary hydrostatic pressure isn't automatically high enough where the transudation threshold is reached is because arterioles constrict, thereby ↑↑ resistance and reducing excessive blood flow through the capillaries.
Hepatic	- Cirrhosis → ↓ hepatic production of albumin → ↓ **intravascular oncotic pressure** → transudation of fluid from systemic veins/venules into interstitium of legs.
Nephrogenic	- Proteinuria → hypoalbuminemia → ↓ **intravascular oncotic pressure** → transudation of fluid from systemic veins/venules into interstitium of legs.
Drugs	- Dihydropyridine calcium channel blockers (i.e., amlodipine, nifedipine). - Imatinib (targets BCR/ABL tyrosine kinase in CML). - Miscellaneous mechanisms not important for USMLE. Just know above drugs do it.
Dietary	- Strict vegetarianism or veganism → ↓ dietary protein consumption → ↓ **intravascular oncotic pressure** → transudation of fluid from systemic veins/venules into interstitium of legs.
Pregnancy	- A little bit of peripheral edema is normal in pregnancy due to compression of IVC.

Hyper-quick causes of unilateral non-pitting edema	
Lymphatic insufficiency	- Malignancy (e.g., peau d'orange of breast), Hx of surgery (e.g., mastectomy), *Wuchereria bancrofti* (elephantiasis).
Thyroid	- Pretibial myxedema (Graves) → mucopolysaccharide deposition in skin + surrounding edema. - Myxedma (severe hypothyroidism) → despite the name, it refers to general "severe hypothyroidism," not just skin changes; can cause carpal tunnel syndrome. - "Pretibial myxedema" is only seen in Graves. Paradoxical hyperthyroidism seen in Hashimoto causing pretibial myxedema is astronomically rare and will get you questions wrong on USMLE.

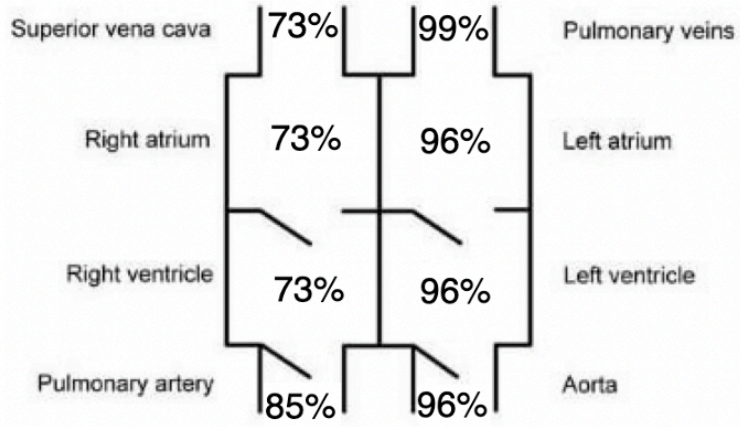
<table>
<tr><th colspan="2">HY Valvular / flow abnormalities on USMLE</th></tr>
<tr><td>Atrial septal defect</td><td>- Fixed splitting of S2
- Can sometimes be associated with a systolic flow murmur, since more blood L → R from the LA → RA means more blood flow across the pulmonic valve. So Q might say “fixed splitting of S2 and a systolic murmur.”
- Sometimes can be seen in Qs as “wide, fixed splitting.” I only mention this because some students get pedantic / ask about this. “Wide splitting” just means right ventricular hypertrophy. So if the Q says “wide, fixed splitting,” they’re saying the patient has RVH due to an ASD.
- Patent foramen ovale = ASD on USMLE. Don’t confuse with patent ductus arteriosus (discussed below).
- USMLE loves asking questions showing you change in oxygen in the chambers of the heart and making you choose ASD, VSD, etc.
 

- For example, you can see above that somehow O2 increases from the SVC to the RA, which is ordinarily impossible. The only way this could occur is if an ASD is present, where oxygenated blood moved from LA → RA.
- ASDs can sometimes be responsible for “paradoxical emboli,” where a DVT leads to stroke. This is ordinarily impossible, since a clot embolizing to the lungs via the venous circulation has no way of reaching the arterial circulation. But if an ASD is present, the clot can go RA → LA → LV → up to the brain, causing stroke.
- 2CK: ASDs do not need to be repaired unless patient has evidence of pulmonary hypertension, RVH, or develops an arrhythmia (i.e., usually AF).
- If adult with ASD has paradoxical embolus (DVT going through the ASD to the brain/arterial circulation), this is also indication for repair.</td></tr>
<tr><td>Ventricular septal defect</td><td>- Holosystolic (aka pan-systolic) murmur at lower left sternal border.
- Can be associated with a diastolic rumble or enlarged left atrium (if more blood going L → R across VSD, then more blood is returning to the LA from the lungs → LA dilatation).
- Seen as part of tetralogy of Fallot (VSD, RVH, overriding aorta, pulmonic stenosis).
- If a VSD is repaired, USMLE wants ↑ LV pressure, ↓ RV pressure, and ↓ LA pressure as the changes now seen in the heart.
- VSD does not cause cyanosis at birth. Only years later after the higher blood flow to the lungs results in pulmonary hypertension, followed by right ventricular hypertrophy and reversal R → L (Eisenmenger) can the patient become cyanotic.
- Murmur can be silent or soft at birth, followed by loud at 7 days of life. The USMLE will ask why the murmur is louder now → answer = decreased pulmonary vascular resistance – i.e., the lungs open up during the first</td></tr>
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	week of life, resulting in decreased RV pressure and an increase in the L → R pressure gradient (louder murmur). - Conversely, if they ask why the murmur was softer at birth compared to now, the answer is "increased pulmonary vascular resistance," where the lungs were still closed at the time, so there was a lesser gradient L → R (softer murmur) at birth. - Similar to ASD Qs, USMLE loves giving you diagrams with changes in O2 between the chambers and then making you infer we have a VSD.  - NBME loves this style of Q. You can see O2 somehow increased from RA to RV. The only way this is possible is if we have a VSD where oxygenated blood moves from LV → RV.  - This one might initially appear a little more difficult. This is Eisenmenger syndrome, where we have a reversal of flow from RV → LV across the VSD. The NBME is known to show this diagram as well. - If you're wondering why oxygen % goes from 99 to 96 from the pulmonary circulation to the LA, this is because of thebesian veins draining the myocardium itself, which open into the different heart chambers, including the LA. If you think that's weird, take it up with NBME, not me, since they have the 99 to 96 drop-down on their diagrams. - 2CK: VSDs are repaired if patient develops pulmonary hypertension, RVH, arrhythmia, Eisenmenger syndrome, recurrent endocarditis (turbulence of blood due to VSD can ↑ risk of valve infections), or aortic regurgitation (if VSD located near the aortic valve).
Atrioventricular septal defect	- Seen in Down syndrome. - Between the atrium and ventricle, aka "endocardial cushion defect," although this latter term can also apply to ASD and VSD in Downs.

Mitral regurgitation	- Holosystolic (pan-systolic) or just regular "systolic," 29 times out of 30. - Q on NBME 20 offline for Step 1 has MR as "mid-systolic," but I contend this is erratum. - Most USMLE questions will not mention it radiating to the axilla. - Highest yield cause of MR on USMLE is post-MI papillary muscle rupture. USMLE is obsessed with this. They'll say hours to days after an MI, patient has new-onset systolic murmur → answer = MR. - Seen **acutely** in rheumatic heart disease (valve scars over years later and becomes mitral stenosis). - Can be caused by general ischemia / dilated cardiomyopathy. - Can cause **JVD** (i.e., back up all the way to the right heart); this is asked multiple times on the new Step 1 NBMEs. - For 2CK: You do not do preoperative stress tests to determine perioperative MI risk if the patient has mere mitral regurg without other risk factors. For example, one of the Surg forms gives a Q where smoker with MR has no shortness of breath or chest pain with exertion, and the answer is "no further management indicated," where exercise stress test is wrong. - If the patient has Sx of heart failure or ischemia, then we do pre-op stress test to determine MI risk. I discuss stress tests later in this chapter. - The mitral valve is replaced if the patient develops severe pulmonary symptoms (i.e., shortness of breath / reduced exercise tolerance), reduced ejection fraction, arrhythmia, or endocarditis if valve function is destroyed.
Mitral stenosis	- Described as "rumbling diastolic murmur with an opening snap"; can also be described as "decrescendo mid-late diastolic murmur" (i.e., following the opening snap). - Can cause a right-sided S4 if the pressure backs up all the way to the right heart (seen on NBMEs sometimes; this confuses students because they think S4 must be LV, but it's not the case). An S4 is a diastolic sound heard in either the LV or RV when there is diastolic stiffening due to high afterload. - 99% of mitral stenoses are due to Hx of rheumatic heart disease (i.e., the patient had rheumatic fever as a child, where at the time it was mitral regurg, but years later it has now become mitral stenosis). - One 2CK NBME Q mentions patient with history of rheumatic heart disease who, years later, now has 4/6 rumbling diastolic murmur **without** an opening snap; this is still mitral stenosis. Although opening snap is buzzy for MS, just be aware it's not mandatory and that this Q exists on NBME. - Other HY presentation on USMLE is pregnant women with new-onset dyspnea in 2nd trimester and a diastolic murmur. This is because 50% increase in plasma volume by 2nd trimester causes the underlying subclinical MS to become symptomatic. Don't confuse this with severe dyspnea and peripheral edema in late third-trimester, which is instead peripartum cardiomyopathy (antibody-mediated). - The 1% of MS that's not due to Hx of RF can be marantic (non-bacterial thrombotic endocarditis; NBTE) → endocarditis seen due to hypercoagulable state in the setting of malignancy, where the vegetations are small and verrucous, on both sides of the valve. This is in contrast to bacterial endocarditis, which causes large, floppy vegetations that lead to MR, not MS. - Libman-Sacks endocarditis seen in SLE is due to antiphospholipid antibodies and is a type of NBTE. - 2CK: Balloon valvuloplasty is the 1st-line Tx for mitral stenosis. This is done if patient has minimal calcification of the valve + has pulmonary HTN. - Mitral valve replacement is done if balloon valvuloplasty fails, if patient has severe MS with dyspnea, arrhythmia, or calcification of the valve.

Mitral valve prolapse	- Most common murmur. - Described as mid-systolic click. - “Myxomatous degeneration” is buzzy term that refers to connective tissue degeneration causing MVP in Marfan and Ehlers-Danlos. - Almost always asymptomatic. On 2CK forms, they want you to know about “mitral valve prolapse syndrome,” which is symptomatic MVP that presents as repeated episodes of “fleeting chest pain” on the left side in an otherwise healthy patient 20s-30s. They might say there is Hx of MI in the family, but this is MVPS, not MI. Answer on surgery form is “no treatment necessary." - USMLE loves using MVP as a distractor in panic disorder questions, particularly on the 2CK Pysch CMS forms. They will give long paragraph about panic attack/disorder + also mention there’s a mid-systolic click; they’ll ask for cause of patient’s presentation → answer = panic disorder, not MVP → student is confused because they say mid-systolic click, but the MVP isn’t the cause of the patient’s presentation; the panic disorder is; MVPs are usually incidental, benign, and asymptomatic. - MVP does not progress to mitral regurg almost always. So don’t think that MVP and MR are the same.
Aortic regurgitation	- Decrescendo holo-diastolic (pan-diastolic) murmur; can also be described as “early diastolic murmur,” or “diastolic murmur loudest after S2.” - Causes wide pulse pressure (i.e., big difference between systolic and diastolic pressures, e.g., 160/50, or 120/40) → results in head-bobbing and bounding pulses (don’t confuse with slow-rising pulses of aortic stenosis). - The bounding pulses can be described on NBME as “brisk upstroke with precipitous downstroke.” In turn, they can just simply say, “the pulses are brisk,” meaning the systolic component is strong. - I would say 4/5 times bounding pulses means AR. The other 1/5 will be PDA and AV fistulae (discussed below). Bounding pulses occur when blood quickly leaves the arterial circulation. In AR, the blood quickly collapses out of the aorta back into the LV. In PDA, it leaves the aorta and enters the ductus arteriosus; in AV fistulae, it leaves for a vein. - Highest yield cause on USMLE is aortic dissection → can retrograde propagate toward the aortic root causing aortic root dilatation and AR. - Even though MVP is most common in Marfan and Ehlers-Danlos, AR is second most common in these patients, since if they get aortic dissection, this can lead to AR. - Can lead to volume overload on the LV and eccentric hypertrophy. - 2CK: Valve is replaced if patient has EF <50%, there is significant left ventricular dilatation, or if severe endocarditis has obliterated the valve.
Aortic stenosis	- Mid-systolic murmur, or just “systolic” murmur; can also be described as “late-peaking systolic murmur with an ejection click.” - Radiates to the carotids. This descriptor shows up quite frequently on NBME (way more than radiation to the axilla for MR). - Causes slow-rising pulses, aka “pulsus parvus et tardus” (don’t confuse with bounding pulses of AR). - SAD → Syncope, Angina, Dyspnea; classic combination seen in AS, albeit not mandatory. If you get a question where they say systolic murmur but you’re not sure of the diagnosis, if they say chest pain or fainting, you know it’s AS. - Often caused by bicuspid aortic valve. The patient need not have Turner syndrome and often won’t. Bicuspid valve is usually inherited as an autosomal dominant familial condition. - The bicuspid valve need not calcify in middle-age prior to the AS forming. Bicuspid valve can present with AS murmur in high schooler, or even a young child, on NBME.

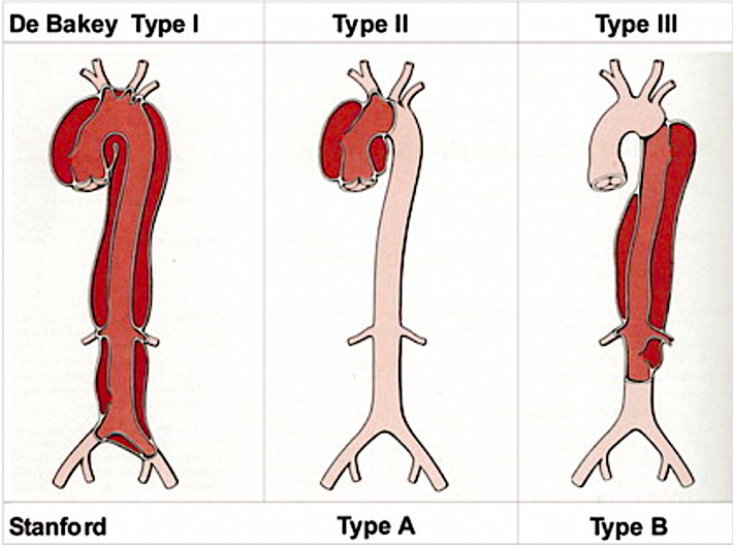
	- For 2CK: Do aortic valve replacement on if 1) cross-section of valve is <1.0 cm^2, or 2) there is SAD. They ask both of these as separate Qs where they want valve replacement.
Tricuspid regurgitation	- Will be described on USMLE as a holosystolic murmur that increases with inspiration. - Right-sided heart murmurs get worse with inspiration → diaphragm moves down → decreased intra-thoracic pressure → increased right-heart filling. - Can cause pulsatile liver. - Highest yield cause of TR on USMLE is **pulmonary hypertension / cor pulmonale.** I see this all over the NBME exams. For whatever reason, these conditions do not cause pulmonic regurg; they cause tricuspid regurg. In other words, if you see tricuspid regurg in a Q, your first thought should be pulmonary hypertension or cor pulmonale (right heart failure due to a pulmonary cause). - IV drug user endocarditis is obvious risk factor for TR, but weirdly nonexistent on USMLE. - Carcinoid syndrome is theoretical cause but lower yield.
Tricuspid stenosis	- Nonexistent murmur on USMLE. I don't think I've ever seen this assessed once on any NBME exam for Steps 1 and 2 combined. - In theory, would be a rumbling diastolic murmur similar to mitral stenosis, but would increase with inspiration since it's on the right side of the heart.
Pulmonic regurgitation	- Same as with tricuspid stenosis, this is a nonexistent murmur on USMLE. I've never seen it assessed. - In theory it would be the same as aortic regurg but on the right (i.e., holo-diastolic murmur), but increases with inspiration.
Pulmonic stenosis	- Seen in tetralogy of Fallot (RVH, VSD, overriding aorta). - Described as mid-systolic murmur, or just regular "systolic" murmur, that increases with inspiration, at the left sternal border, 2nd intercostal space. This is the theoretical location, whereas AS is the 2nd intercostal space on the right, not left. But the USMLE often isn't strict about murmur locations this way.
Patent ductus arteriosus (PDA)	- Ductus arteriosus is special vessel in fetal circulation that connects the proximal pulmonary trunk to the descending arch of the aorta. This allows for blood to bypass the high-resistance lungs *in utero*. After birth, this vessel should close, resulting in a remnant called the ligamentum arteriosum, but sometimes it does not close → PDA. - USMLE wants you to know PDA is an "extra-cardiac L→R shunt." - If a PDA occurs, blood moves in the neonate L→R (i.e., opposite of *in utero)* from the descending arch of the aorta to the pulmonary trunk.  - You can see in the above diagram, somehow the blood become more oxygenated from the RV to the pulmonary artery, which is ordinarily

	impossible. The only way this could have occurred is if oxygenated blood came L→R from the aorta to the pulmonary artery via a PDA. - Murmur described three ways on USMLE: 1) continuous, machinery-like murmur; 2) pan-systolic pan-diastolic murmur (meaning it's continuous throughout both systolic and diastole); and 3) to-and-fro. The latter shows up on 2CK offline NBME 6. - Classically associated with congenital rubella (HY). They'll give a kid born with a PDA and then ask what the mom experienced while pregnant; answer = arthritis and/or rash (rubella often presents as arthritis in adults). - Indomethacin (NSAID) will close the PDA. - Prostaglandin E1 is used to keep a PDA open (if a kid with congenital heart malformations is born cyanotic and we need to buy time until surgery). - A PDA can mask cyanosis in a newborn in a variety of conditions (i.e., hypoplastic left heart syndrome or pre-ductal coarctation). If they tell you a kid is born with normal APGAR scores but a week later becomes cyanotic and they ask why, the answer is "closure of ductus arteriosus."
Tetralogy of Fallot	- 1) Pulmonic stenosis; 2) RVH; 3) overriding aorta; 4) VSD. - If you're asked which component most determines prognosis, the answer is the degree of pulmonic stenosis. - The child will not be cyanotic at birth, but then years later, will develop Eisenmenger syndrome (i.e., a reversal of the L→R shunt over the VSD to be R→L) and cyanosis, where the stem gives a school-age kid who squats on the playground to relieve symptoms. - Squatting ↑ afterload, which ↑ LV pressure, which ↓ the pressure gradient of the R→L shunt, thereby mitigating cyanosis. - Squatting also ↑ preload by ↑ venous return back to the right heart. But it is the effect of ↑ afterload that is most related to the ↓ in symptoms. - For 2CK: Tx is surgical correction in infancy or early childhood.
Functional (flow) murmur	- Systolic murmur seen in the setting of higher heart rate caused by infection, anemia, or pregnancy. Caused by increased flow across the pulmonic and/or aortic valves. - Known as a functional murmur because this means it goes away once the heart rate comes back down. - Seen all over 2CK Peds forms, where they try to trick you into thinking the kid has a valvular pathology of some kind, but there isn't; there will merely be an infection or simple viral infection. - Can be seen sometimes with ASD, where the patient will have fixed splitting of S2 "plus a systolic murmur" → merely higher right-sided volume, so more flow across the pulmonic valve.
Venous hum	- On 2CK Peds form; described as a murmur in the neck that abates when the kid is laid supine + the neck rotated. - Benign + don't treat.
"Ball-in-valve" murmur	- Associated with cardiac tumors (i.e., myxoma in adult, or rhabdomyoma in kids for tuberous sclerosis). - Described as a diastolic rumbling murmur (and rarely systolic) that abates when the patient is re-positioned unconventionally (e.g., onto his or her right side).
Persistent fetal hypertension	- Aka persistent fetal circulation. - Q will give a post-term birth at 42 or 43 weeks + meconium-stained fluid + echo of the neonate shows a R→L shunt across the foramen ovale. Student says, "Wait, but isn't the foramen ovale between the atria, and that's only open in the fetus but is supposed to close after birth?" Correct. Hence we have *persistent* fetal circulation. - Post-term birth is risk factor for meconium aspiration syndrome, which in turn is a risk factor for persistent fetal circulation.

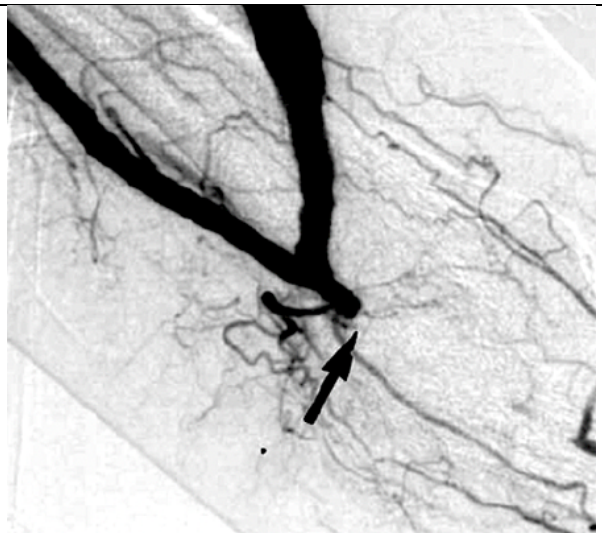
	- Answer on USMLE will be "failure of pulmonary vasodilation." Meconium aspiration syndrome can ↓ opening of the lung vasculature, leading to ↑ right heart pressure and ↑ risk of persistent fetal circulation.
S3 versus S4	- Both are diastolic sounds. - S3 is due to high volume/preload in the left ventricle, causing a reverberation against the wall. - S3 can sometimes be physiologic (i.e., normal / no problem) in pregnancy and high-endurance athletes. Patient will have eccentric hypertrophy (sarcomeres laid in linear sequence). If pathologic, it is due to dilated cardiomyopathy with reduced ejection fraction (<55%) or high-output cardiac failure (EF >70%). - There is one question on IM CMS form 7 where they give an S3 in diastolic dysfunction. I'm convinced this is an erratum, but I need to mention it because it exists on the NBME form. - S4 is due to high pressure/afterload on the left (but sometimes right) ventricle, causing a stiffened ventricle with diastolic dysfunction and concentric hypertrophy (sarcomeres laid in parallel). It is always pathologic. It is usually caused by systemic hypertension causing afterload on the LV, or aortic stenosis. - S4 can sometimes be right-sided on USMLE. There is a 2CK Q where they give severe mitral stenosis and say there's an S4, but it's for the RV not LV. Some weird/annoying points: - The combo of S3 and S4, seen together in the same vignette, can be seen in high-output cardiac failure. For example, they will say a patient has an AV fistula/conduit or Paget disease, and they will say there's S3 and S4 and ask for diagnosis → answer = high-output cardiac failure. The take-home point is that high-output failure can present with either an isolated S3 or the combo of S3 and S4 together, but never S4 alone on USMLE.
Loud P2	- One of the highest yield cardiac sounds on USMLE, almost always overlooked by students. - Means pulmonary hypertension or cor pulmonale on USMLE. - The pulmonic valve slams shut due to high pressure distal to it. - For example, they'll give a smoker who simply has a loud P2 → this just means patient has pulmonary hypertension. Not complicated. - Also recall that I said above that highest yield cause of tricuspid regurg on USMLE is pulmonary hypertension / cor pulmonale. So I want you to remember both TR and loud P2 for this. - Sometimes the UMSLE will just say "loud pulmonic component of S2," or "loud S2," rather than saying "loud P2." I've never seen "loud A2" on USMLE, but in theory this means systemic hypertension. - A soft P2 refers to pulmonic stenosis, but is LY.
Wide splitting of S2	- Means right ventricular hypertrophy on USMLE. - A2 and P2 are far apart. - You don't have to worry about the mechanism. But in short, the more pressure you have in a ventricle, the more delayed the closure of the semilunar valve. So if we have RVH, P2 occurs later, widening the split. - Wide splitting of S2, right-axis deviation on ECG, and right bundle branch block (RBBB) all = right ventricular hypertrophy on USMLE.
Paradoxical splitting of S2	- Means left ventricular hypertrophy on USMLE. - A2 occurs after P2 (normally we have A2 before P2). - Left ventricular pressure is high and A2 delayed to the point that it actually occurs on the opposite side of P2. - Paradoxical splitting of S2, left-axis deviation on ECG, and left bundle branch block (LBBB) all = left ventricular hypertrophy on USMLE.
Coarctation of the aorta	- Refers to narrowing of the aortic arch (this is referred to as coarctation; do not use the word stenosis to describe this).

	- Classically seen in Turner syndrome, but absolutely not mandatory. Shows up idiopathically in plenty of NBME Qs. I point this out because students often think the patient must have Turner syndrome. - Presents as upper extremities that have higher BP, brisk pulses, and are warmer; the lower extremities have lower BP, weak pulses, and are cooler. - Sometimes the Q can just say, "the radial pulses are brisk." → The implication is, "Well if they're saying specifically that the radial pulses are brisk, that must mean the pulses in the legs aren't." - Murmur sound not important for USMLE. Can sometimes be described as a systolic murmur heart in the infrascapular region. - Can cause LVH with left-axis deviation ECG (on Step 1 NBME). - USMLE doesn't give a fuck about pre- vs post-ductal. Pre-ductal in theory will be a very sick neonate. Post-ductal will be an adult (most cases).
Subclavian steal syndrome	- Confusing condition when you're first learning things that is low-yield on Step 1 but high-yield on 2CK. - The vertebral artery (goes to brain) is the first branch of the subclavian artery (goes to arm). - If there is a narrowing/stenosis of the proximal subclavian prior to the branch point of the vertebral artery, this can lead to lower pressure in the vertebral artery. - This can cause a backflow of blood in the vertebral artery, producing miscellaneous neuro findings such as dizziness. - **Blood pressure is different between the two arms.** - USMLE will ask the Q one of two ways: 1) they'll give you dizziness in someone who has BP different between the arms and then ask for merely "subclavian steal syndrome," or "backflow in a vertebral artery" as the answer. Or 2) they'll give you BP in one of the arms + give you dizziness, then the answer will be, "Check blood pressure in other arm." - Next best step in Dx is CT or MR angiography (asked on 2CK NBME). - I should point out that probably 3/4 questions on USMLE where blood pressure is different between the arms, this refers to aortic dissection. But 1/4 is subclavian steal syndrome. As per my observation.
Vertebral artery stenosis	- Presents same as subclavian steal syndrome with otherwise unexplained dizziness, but blood pressure is not different between the arms because the subclavian is not affected. - Caused by atherosclerosis. CT or MR angiography can diagnose. - "Vertebrobasilar insufficiency" is a broader term that refers to patients who have either subclavian steal syndrome or vertebral artery stenosis.
Vertebral artery dissection	- 2CK Neuro forms assess vertebral artery dissection, where they want you to know a false lumen created by dissection in a vertebral artery can lead to stasis and clot formation, which in turn can embolize to the brain and cause stroke. - NBME can mention recent visit to a chiropractor (neck manipulation is known cause). - The answer on the NBME is **heparin** for patients who have experienced posterior stroke due to vertebral artery dissection. Sounds weird because it's arterial, but it's what USMLE wants. Strokes of the vertebral artery tend to cause lateral medullary syndrome (ipsilateral Horner syndrome + dysphagia).
Carotid artery dissection	- Shows up on 2CK form as patient with stroke-like presentation + who simultaneously has **ipsilateral facial/neck pain.** - The pain is due to stretching of nociceptors secondary to vascular dilation. - Stasis within false lumen can lead to embolus to brain/eye.
Carotid artery stenosis **(exceedingly HY for 2CK/3)**	- Caused by atherosclerosis.

	- HTN biggest risk factor for atherosclerosis specifically of the carotids (strong systolic impulse pounds the carotids → endothelial damage → atheromatous plaque formation). - Carotid bruit only seen in about 25% of Qs. Don't rely on this as crutch. - Vignette will give a stroke, TIA, or retinal artery occlusion in the setting of a patient with HTN. → You have to be able to make the association that a plaque from one of the carotids has launched off, since HTN = ↑ risk. - USMLE will then ask for management (2CK only): - Do carotid duplex ultrasonography as next best step in diagnosis to look for degree of occlusion. I've never seen carotid angiography as a correct answer on NBME exams. - If occlusion >70% symptomatic, or >80% asymptomatic, then do endarterectomy. "Symptomatic" = stroke, TIA, or retinal artery occlusion. A mere bruit is not a symptom; that is a sign. - If under these thresholds, do medical management only, which requires a triad of: 1) statin; 2) ACEi or ARB; and 3) anti-platelet therapy. - The USMLE will not force you to choose between low- and high-potency statins. - USMLE tends to list lisinopril as their favorite ACEi for HTN control. - It's to my observation aspirin alone is sufficient on NBME exams for anti-platelet therapy, even though in real life patient can receive either aspirin alone; the combo of aspirin + dipyridamole; or clopidogrel alone. - USMLE will not give borderline carotid occlusion thresholds – i.e., they'll say either 30% or 90%. If they list the % as low, look at the vignette for the drugs they list the patient on. Sometimes they'll show the patient is already on statin, lisinopril, and aspirin, and then the answer is just "continue current regimen." I have once seen "add clopidogrel" as a wrong answer in this setting, which makes sense, since the combo of aspirin + clopidogrel is never given anyway. - Sometimes they will give you a low carotid occlusion % + say the patient is on 2 of 3 drugs in the triad, and then the answer is just "add aspirin," or "add statin," or "add lisinopril." - If the vignette doesn't mention elevated BP but says you have some random dude over 50 with a stroke, TIA, or retinal artery occlusion, the next best step is carotid ultrasonography to look for carotid stenosis. In other words, it is assumed the patient has a carotid plaque in this setting. - If the vignette gives patient with episodes of unexplained *syncope* or light-headedness, but not stroke, TIA, or retinal artery occlusion, then the next best step is ECG, followed by Holter monitor, looking for atrial fibrillation (AF causes LA mural thrombus that launched off to brain/eye). - The triad of 1) statin; 2) ACEi or ARB; and 3) anti-platelet therapy is also done for general peripheral vascular disease unrelated to carotid stenosis (i.e., if a patient has intermittent claudication). - Stroke, TIA, or retinal artery occlusion, if they don't mention HTN, but they mention an *abdominal* bruit, you will still do a carotid duplex ultrasound. The implication is that the bruit in the abdomen could be a AAA or RAS, where atherosclerosis in one location means atherosclerosis everywhere, so the patient likely has carotid stenosis by extension. They once again need not mention carotid bruit; apparently it is not a sensitive finding (i.e., we cannot rule-out ↑ occlusion just because we don't hear it).
Aortic dissection	- Classically presents as severe upper chest pain radiating to the back between the scapulae. - As discussed above in the aortic regurg section, USMLE loves this as most common cause of AR due to retrograde propagation toward the aortic root. For example, patient with Hx of HTN, cocaine use, or a connective

	tissue disorder (i.e., Marfan, Ehlers-Danlos) who has a diastolic murmur, you should be thinking immediately that this is dissection. - "Medial necrosis" is a term that is used on NBME exams to describe changes to the aorta in dissection. In the past, "cystic medial necrosis" used to be buzzy for dissection due to Marfan syndrome, but I haven't seen USMLE care about this. I have, however, seen a dissection Q on NBME where it is due to hypertension, and simply "medial necrosis" is the answer. - As mentioned above, 3/4 Qs where BP is different between the arms refers to aortic dissection. A Q on 2CK IM form 7 has "thoracic aortic dissection" where not only is the BP different between the arms, but it's also different between the L and R *legs* (i.e., L-leg BP is different from R-leg BP) → sometimes thoracic aortic dissections can anterograde propagate all the way down to the abdominal aorta.  - You do **not** need to memorize these aortic aneurysm types. I'm just showing you that if the common iliacs are involved (as with left image), BP can differ as well *between the legs*. For 2CK: - CXR is done first to view "widening of the mediastinum." The next step is "CT of the chest," or "CT angiography." They refer to same thing in this setting. If not listed, choose "aortic angiography" and "aortography." - Tx for ascending aortic aneurysm (type A) = labetalol + surgery. - Tx for descending aortic aneurysm (type B) = labetalol alone initially.
Traumatic rupture of the aorta	- Caused by deceleration injury. Most common cause of death due to car accident or fall. Exceedingly HY on 2CK. - Will be described as patient following an MVA who has "widening of the mediastinum" on CXR. Next best step is CT of chest or CT angiography. If not listed, choose aortography or aortic angiography. - Labetalol used first-line in aortic dissection and traumatic rupture of the aorta. If they force you to choose another drug after, nitroprusside is the answer on NBME. - Labetalol is answer on NBME even in patient who has *low BP* due to rupture or dissection due to the drug ↓ shearing forces. I've seen students get this wrong saying, "But patient has low BP though." My response is, file a complaint with the exam not with me. - 2CK Q gives "esmolol + nitroprusside" as answer to a traumatic rupture Q, but almost always, they will just want "labetalol." - Emergency surgical repair is indicated following IV drug administration.
Aortic aneurysm	- Can present as "visible pulsation" on USMLE. - For aortic aneurysm, they can say "visible pulsation above the manubrium," or "pulsatile mass above the manubrium." **There can also be**

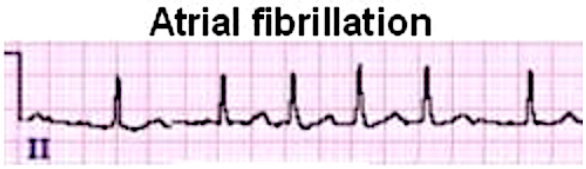
	a tracheal shift. I've seen students select pneumothorax here. But for whatever reason you can get tracheal shift in thoracic aortic aneurysm. For AAA, there can be "visible pulsation in the epigastrium." - Biggest risk factor for AAA is smoking. - For Family Med, do a one-off abdominal ultrasound in men 65+ who are ever-smokers. - For Surgery, AAA repair is indicated if the aneurysm is >5.5 cm or the rate of change of size increase is >0.5cm over a 6-month period. This is on Surgery form, where they give a patient with a 4-cm AAA and ask why serial ultrasounds are indicated → answer = "size of aneurysm." - Don't do AAA repair on USMLE in patient who has advanced comorbidities or terminal disease, e.g., stage 4 lung cancer. - In general, perioperative MI risk is assessed using a pre-op stress test. 2CK NBME Q has dipyridamole and thallium pharmacologic stress test as answer in patient with 6-cm AAA prior to surgery. - Diabetes is protective against aneurysm. Non-enzymatic glycosylation of endothelium causes stiffening of the vascular wall. - Tangential: 2CK Surg loves "pulsatile hematoma" in the neck in trauma patients, where the answer is "endotracheal intubation." Sounds nitpicky, but shows up repeatedly.
Arteriovenous fistula	- Can be idiopathic, iatrogenic (i.e., dialysis), from injury (i.e., stab wound), or caused by other disease (i.e., hereditary hemorrhagic telangiectasia or Paget disease of bone). - Similar to aortic aneurysms, AV fistulae can sometimes present with pulsatile mass, *but in a weird location*, e.g., around the left ear in patient with tinnitus (on NBME exam). Student says, "Why is it at the left ear though?" → No fucking idea. Take it up with NBME. - Highest yield point is they can cause **high-output cardiac failure.** This is because blood quickly enters the venous circulation from the arterial circulation → combo of ↑ preload back to right heart + poorer arterial perfusion distal to the fistula → compensatory ↑ CO. - AV fistulae can sometimes present with a **continuous machinery murmur similar to a PDA,** since blood is continuously flowing through it. They might say a continuous machinery-like murmur is auscultated in the leg at site of prior stab wound. - As discussed earlier, they can present with bounding pulses similar to AR. - Student says, "Well how am I supposed to know if it's AV fistula then if it sounds like other conditions too?" → by paying attention to HY points like, "Is there lone S3 or S3/4 combo or EF >70%? Is there Hx of penetrating trauma? Or does the patient have Paget? Etc." - 2CK NBME Q shows you obscure angiogram of a fistula in the leg + tells you there's a continuous machinery murmur; they ask what most likely determines prognosis in this patient → answer = "size of lesion."

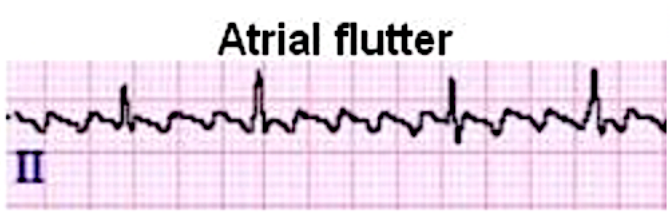


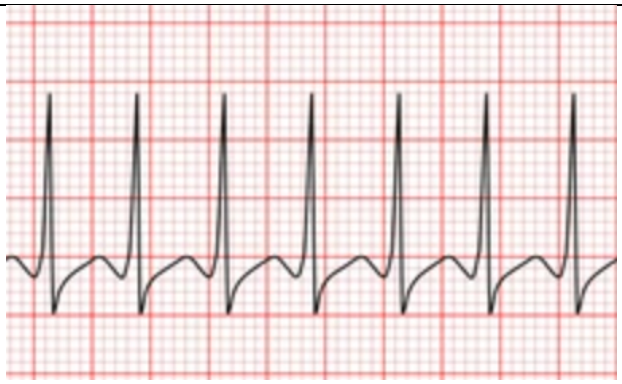
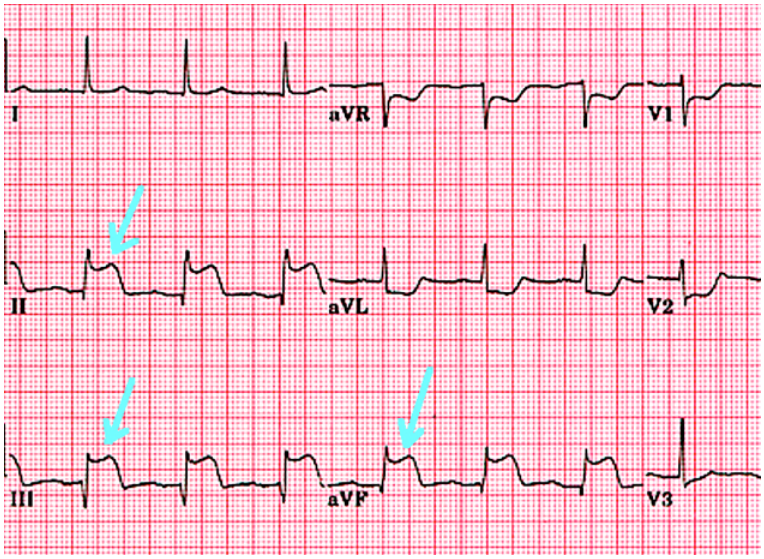
- NBME exam shows obscure image similar to above (without the arrow) + they tell you there's continuous murmur → answer = "size of lesion."
- Another NBME Q gives 45-year-old male will nosebleeds since adolescence + S3 heart sound + dyspnea + they show you pic of tongue; they ask for the cause of dyspnea.



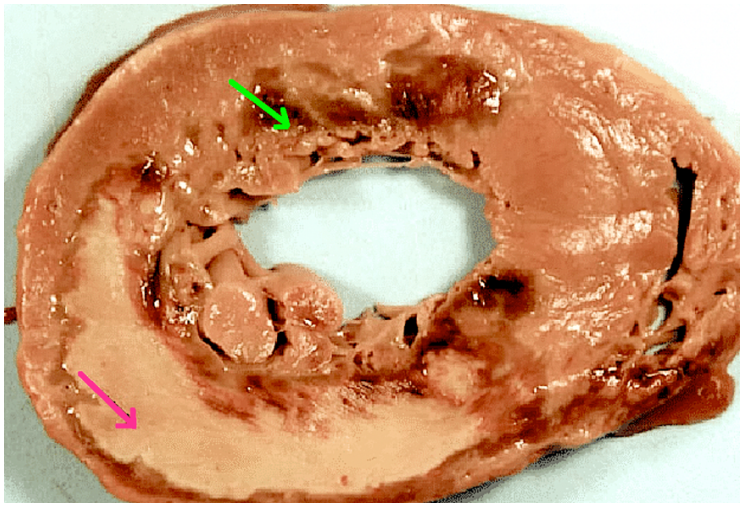
- Answer = "Pulmonary arteriovenous fistula" (leading to high-output failure); diagnosis is hereditary hemorrhagic telangiectasia. USMLE will basically always show you a pic of red dots on the tongue/mouth or finger in a patient with nosebleeds.
- Likewise, be aware intraosseous AV fistulae can occur in Paget, as mentioned before.

HY Murmur / ECG points for USMLE	
Atrial fibrillation (AF)	- Described as "irregularly irregular" rhythm with absent p-waves.  - Notice how the QRS complexes are at random and irregular distances from one another. This is the "irregularly irregular" pattern. - AF is hugely important because it can cause turbulence/stasis within the left atrium that leads to a LA mural thrombus formation. This thrombus can launch off (i.e., become an embolus) and go to brain (stroke, TIA, retinal artery occlusion), SMA/IMA (acute mesenteric ischemia), and legs (acute limb ischemia). These vignettes are higher yield for 2CK, but the concept is important for Step 1. - AF HY in older patients, especially over 75. Vignette will usually be an older patient with a stroke, TIA, or retinal artery occlusion, who has *normal* blood pressure (this implies carotid stenosis is not the etiology for the embolus). - AF usually is paroxysmal, which means it comes and goes. The vignette might say the patient is 75 + had a TIA + BP normal + ECG shows sinus rhythm with no abnormalities → **next best step is Holter monitor (24-hour ambulatory ECG monitor)** to pick up the paroxysmal AF (e.g., when the patient goes home and has dinner). - After AF is diagnosed with regular ECG or Holter, 2CK wants **echocardiography** as the next best step to visualize the LA mural thrombus. - Patient who has severe abdominal pain in setting of AF or hyperthyroidism (which can cause AF), diagnosis is acute mesenteric ischemia; next best step is mesenteric angiography; Tx is laparotomy if unstable (answer on NBME). - Severe pain in a leg + absent pulses in patient with irregularly irregular rhythm = acute limb ischemia; USMLE wants "embolectomy" as answer. - Any structural abnormality of the heart, either due to LV hypertrophy, ischemia, growth hormone/anabolic steroid use, prior MI, etc., can lead to AF. - For 2CK, you need to know AF patient will get either NOAC or warfarin. Aspirin is no longer indicated for AF management. This is determined by the CHADS2 score. There are variations of the score, but the simple **CHADS2** suffices for USMLE → **C**HF, **H**TN, **A**ge 75+, **D**iabetes, **S**troke/TIA/emboli. Each component is 1 point, but stroke/TIA/emboli is 2 points. More points favors warfarin. - "Emboli" refers to Hx of AF leading to stroke, TIA, acute, mesenteric ischemia, or acute limb ischemia – i.e., any Hx of embolic event. 2CK IM form 7 gives short vignette of 67F with chronic AF + Hx of acute limb ischemia + no other info relating to CHADS, and answer is warfarin to prevent recurrence; aspirin is wrong. - Some students will ask about NOACs, e.g., apixaban, etc., for non-valvular AF → I've never seen NBME care about this stuff. They seem to be pretty old-school and just have warfarin as the answer, probably because there isn't debate around whether it can be used; use of NOACs is less textbook. - AF patient should also be on rate control before rhythm control. The USMLE actually doesn't give a fuck about this component of management, although in theory metoprolol or verapamil is standard. You could be aware for Step 3 that flecainide is first-line for rhythm control if patients fail rate-control and have a structurally normal heart and no coronary artery disease. - NBME 9 for 2CK has "electrical cardioversion" as the answer for patient with AF who has hemodynamic instability (i.e., low BP). What you need to know is: sometimes AF can trigger "rapid ventricular response," where HR goes >150 and low BP can occur.
Atrial flutter	- Has classic sawtooth appearance.

	 - Low yield for USMLE. I think it's asked once on a 2CK NBME. But as student you should know it exists / the basic ECG above.
Ventricular tachycardia (VT)	- Causes wide-complex QRS complexes (>120 ms; normal is 80-120 ms).  - Exceedingly HY for 2CK that you know VT is wide-complex, whereas SVT is narrow-complex. If you look at above ECG, even if you say, "No idea what I'm looking at." You can tell the complexes look wide like mountains in comparison to a typical ECG. - VT is treated with anti-arrhythmics – i.e., amiodarone. If patient has coma or hemodynamic instability (low BP), the NBME answer is direct current countershock or cardioversion (same thing). - Premature ventricular complex (PVC) is asked on 2CK.  - Note on the above strip, we have a wide complex (meaning ventricular in origin) that occurs earlier (hence premature). What they do on the NBME is show you this strip and ask where this abnormality originates from, then the answer is just "ventricle." - Don't treat PVCs on USMLE.
Supraventricular tachycardia (SVT)	- Causes narrow / needle-shaped complexes. Make sure you're able to contrast this with VT above, which is wide-complex.

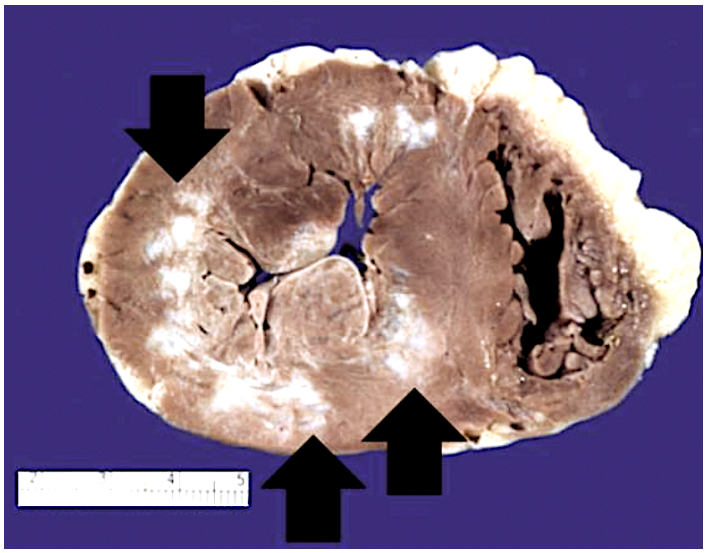
	 - Notice the complexes are narrow / look like needles. This means the tachy originates above the ventricles (hence SVT). - Treatment of SVT exceedingly HY on 2CK. - First step is carotid massage (aka vagal maneuvers). In pediatrics, they can do icepack to the face. - If the above doesn't work, the next step is give adenosine (not amiodarone). - Same as with VT, if the patient has coma or low BP, shocking the patient is the first step. In other words, for both SVT and VT, you must shock first in the setting of coma or hemodynamic instability. It's for stable SVT and VT that the treatments differ on USMLE.
Acute MI (STEMI)	- Will present as ST-elevations in 3-4 contiguous leads.  - The above is an inferior MI, as evidenced by ST-elevations in leads II, III, and aVF. The answer for the affected vessel is the posterior descending artery (PDA supplies the diaphragmatic surface of the heart); since >85% of people have right-dominant circulation (meaning the PDA comes of the right main coronary), sometimes the answer for inferior MI can just be "right coronary artery." - If the Q says left-dominant circulation, the sequence USMLE wants is: left main coronary → left circumflex → PDA. - The apex of the heart is supplied by the left anterior descending artery (LAD). If there are ST-elevations in leads V1-V3, choose LAD as the answer. - The left-lateral heart is supplied by the left circumflex artery. If there are ST-elevations in leads V4-V6 for lateral MI, choose left circumflex. - Reciprocal ST-depressions in the anterior leads V1-V3 can reflex posterior wall MI (i.e., we have "elevations" out the back of the heart, so they look like depressions on the anterior wall leads).

- USMLE wants you to know gross appearance of fresh vs several-day-old myocardial infarcts:



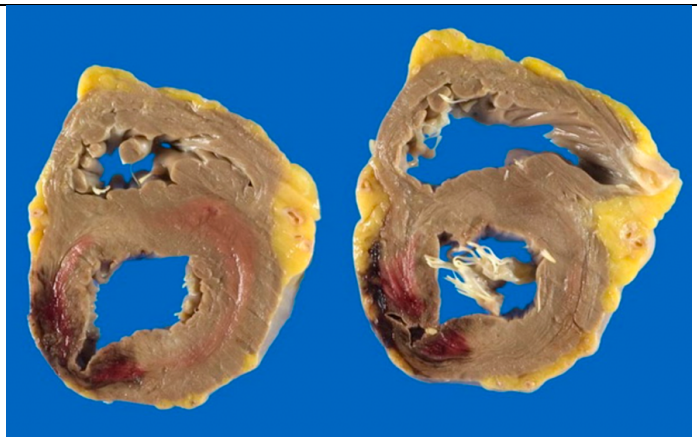
At 12-24 hours, fresh infarcts show dark mottling (green arrow); by 10-14 days, an infarct becomes a yellow, softened area (**pink arrow**).

- Old infarcts will appear white. There is an NBME Q that shows image similar to the following, where the answer is “congestive heart failure resulting from repeated infarcts”:



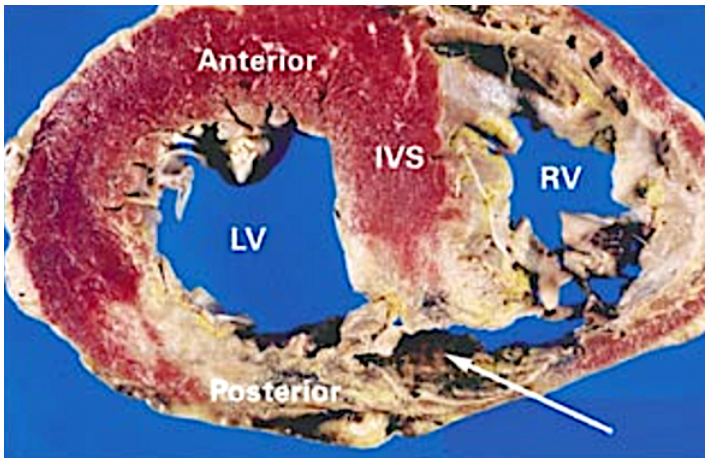
Note the old infarcts are white in appearance (black arrows).

- Left ventricular free-wall rupture can cause tamponade (Beck triad):



Note the myocardial free-wall rupture at the lower left aspect.

- Ventricular septal rupture has similar image:

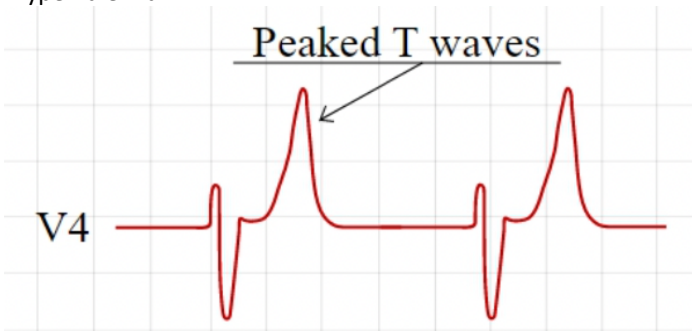


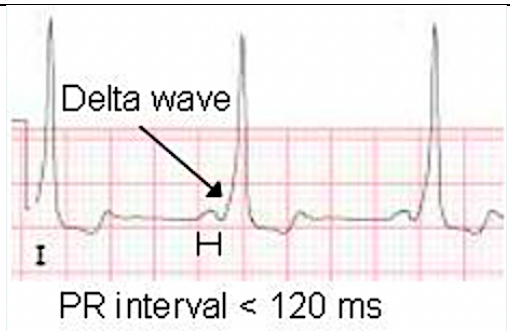
Note the thinned/ruptured area of the interventricular septum (white arrow).

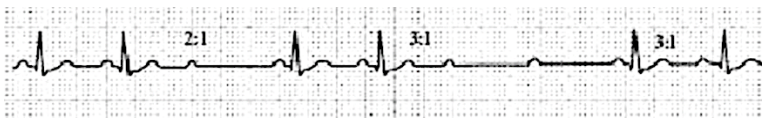
- Post-MI papillary muscle rupture resulting in mitral regurg is exceedingly HY. As discussed earlier, if patient has MI followed by new-onset systolic murmur hours to days later, with or without dyspnea, that's mitral regurg.
- Stroke-like presentation in patient who had MI weeks ago → "embolus from ventricular septal aneurysm" (on 2CK Neuro form).
- Most common cause of death due to MI is ventricular fibrillation (VF).
- Fibrosis of myocardium in the months-years post-MI increases risk of arrhythmias such as AF, SVT, VT, etc. There's no specific arrhythmia you need to memorize. Just know the risk is there in the future.
- Q waves on an ECG mean old MI / history of MI. The vignette might give you patient who has light-headedness / fainting + they say patient has Q waves in II, III, aVF, and the answer will be something like "paroxysmal supraventricular tachycardia." Student thinks this specific arrhythmia matters, but it doesn't. The point is that Hx of MI means patient is at risk for nearly any arrhythmia now.
- MI classically causes coagulative necrosis of the myocardium.
- With cardiogenic shock as a result of MI, the arrows USMLE wants are: ↓ cardiac output, ↑ peripheral vascular resistance, ↑ PCWP.
- MI can lead to acute tubular necrosis from cardiogenic shock → acute drop in renal perfusion. This is not pre-renal. I discuss this in detail in the renal section.
- First treatment for MI is aspirin. After aspirin is given, the next drug to give is clopidogrel (an ADP P2Y12 blocker) as dual anti-platelet therapy.

	- USMLE wants you to know anyone with acute coronary syndrome (i.e., MI or unstable angina) gets cardiac catheterization. This is answer on new 2CK NBME exam. - It's to my observation that more extensive management of MI on USMLE, such as use of beta-blockers, nitrates, morphine, oxygen, statin, percutaneous coronary intervention, etc., isn't assessed in detail. I can comment, however, that one 2CK Q wants you to know nitrates are contraindicated in right-heart MIs, which includes inferior MI in most people due to the right coronary supplying the PDA. This is because right-sided MIs are preload-dependent, which means they need sufficient preload to maintain BP. - 2CK: Percutaneous coronary intervention (PCI) is done in patients with STEMIs within 90 minutes of reaching hospital.
Pericarditis	- Shows up on ECG as diffuse ST-elevations (i.e., in all leads rather than 3-4 contiguous leads as with MI). PR depressions can also be seen, but I've never seen the USMLE give a fuck about the latter. - Patient will have pain that's worse when lying back, better when leaning forward. In turn, the patient can present walking through the door bent over at the waist. - Serous pericarditis will be post-viral, secondary to autoimmune disease, or due to cocaine use. - Step 1 NBME Q gives pericarditis + a bunch of different organism types (i.e., bacterium, fungus, etc.), and answer is "virus." - Patient with rheumatoid arthritis or SLE notably at risk for pericarditis. In other words, don't get confused if they mention pericardial friction rub in vignette of RA or SLE; this is common. - For cocaine use, they'll say a 22-year-old male has chest pain after a night of heavy partying + ECG shows diffuse ST-elevations → Dx = pericarditis. - Fibrinous pericarditis is post-MI and occurs as two types: 1) literally "post-MI fibrinous pericarditis," which will simply be friction rub within days of an MI; 2) Dressler syndrome (antibody-mediated fibrinous pericarditis occurring 2-6 weeks post-MI). - Uremic pericarditis is HY for 2CK. Q will give ultra-high creatinine and BUN and say there's a friction rub → treatment = hemodialysis. - ECG is first step in Dx of pericarditis, but USMLE wants echocardiography as next best step in order to visualize a concomitant effusion that can occur sometimes. Vignette will give you stereotypical pericarditis + will ask for next best step in diagnosis; ECG might not be listed and you're like huh? → Answer is echocardiography to look for potential effusion concomitant to the pericarditis. - Treatment for pericarditis is same as acute gout → NSAIDs, colchicine, steroids.
Chronic constrictive pericarditis	- I should make note that chronic constrictive pericarditis is a separate condition that doesn't present with the standard pericarditis findings as described above. - This is low-yield for USMLE, but students ask about it because it can be confused with tamponade. - There's two ways this can show up: 1) **Tuberculosis** is a classic cause; there may or may not be calcification around the heart on imaging. So if you get a Q where patient has TB + some sort of heart-filling impairment → answer = chronic constrictive pericarditis. 2) **Kussmaul sign** will be seen in the Q, where JVD occurs with *inspiration* rather than expiration. - Normally, inspiration facilitates RA filling (↓ intrathoracic pressure → ↑ pulmonary vascular compliance/stretching → ↑ high-low pressure gradient from right heart to the lungs → ↓ in afterload on RV from the lungs → blood moves easier from right heart to the lungs → blood is pulled easier from SVC/IVC to the RA).

	- However, if there is ↑ compressive force on the heart, the ↑ in negative intrathoracic pressure during inspiration is not transmitted to the right side of the heart, so JVP does not ↓ (and can even paradoxically ↑). - In tamponade, however, as discussed below, the ↑ in negative intrathoracic pressure during inspiration *is able* to be transmitted to the right side of the heart, so Kussmaul sign does not occur. This is likely because in constrictive pericarditis, the rigid pericardium prevents expansion of the right heart altogether, whereas in tamponade, the pericardium isn't rigid per se, but is just filled with blood that can move/shift during the respiratory cycle, thereby allowing right heart expansion during inspiration.
Pericardial effusion / Cardiac tamponade	- Cardiac tamponade = pericardial effusion + low blood pressure. - What determines whether we have a tamponade or not is the *rate* of accumulation of the fluid, not the volume of the fluid – i.e., a stab wound or post-MI LV free-wall rupture resulting in fast blood accumulation, even if smaller volume, might cause tamponade, but cancer resulting in slow, but large, accumulation might not cause tamponade. - Tamponade presents as **Beck triad: 1) hypotension, 2) JVD, 3) muffled/distant heart sounds.** The question will basically always give hypotension and JVD. Occasionally they might not mention the heart sounds. But you need to memorize Beck triad as HY for tamponade. - Pulsus paradoxus (i.e., drop in systolic BP >10 mm Hg with inspiration) is classically associated with tamponade, although not frequently mentioned in vignettes. I've seen a 2CK NBME Q where they say "the pulsus paradoxus is <10 mm Hg," which is their way of saying the Dx is not tamponade. I consider that wording odd, but it's what the vignette says. - ECG will show electrical alternans / low-voltage QRS complexes.  - You can see the amplitudes (i.e., heights) of the complexes are short. This refers to "low-voltage." You can also see the heights ever so slightly oscillate up and down. This refers to electrical alternans. They show this ECG twice on 2CK NBMEs. - If the Q asks for next best step in *diagnosis,* choose ECG as first step if listed. If not listed, then choose echocardiography, which confirms fluid over the heart. - If the Q asks for next best step in management for tamponade when the vignette is obvious, choose pericardiocentesis or pericardial window. USMLE will not list both; it will be one or the other. NBME 8 offline for 2CK has pericardial window as answer, where pericardiocentesis isn't listed.
Torsades de pointes (TdP)	- HY type of VT that has sinusoidal pattern on ECG.

	 - USMLE wants you to know this can be caused by some anti-arrhythmic agents, such as the sodium- and potassium-channel blockers, such as quinidine and ibutilide, respectively. They ask this directly on the NBME exam, where Q will say patient is given ibutilide + what is he now at increased risk of → answer = torsades. - QT prolongation is risk factor for development of TdP. Agents such as TCAs, macrolides, anti-psychotics, and metoclopramide prolong the QT. - Tx USMLE wants is magnesium (asked directly on new 2CK form), which stabilizes the myocardium in TdP.
Peaked T wave	- Seen in hyperkalemia.  - Asked once on one of the 2CK forms, where they show the ECG. - Highest yield point is that if a patient has hyperkalemia and ECG changes, the Tx USMLE wants is IV calcium gluconate or calcium chloride, which stabilizes the myocardium. Calcium gluconate is classic, but calcium chloride shows up as an answer on a 2CK NBME.
U-wave	- Means hypokalemia.  - Shows up on NBME for 2CK in anorexia patient. First time I've ever seen it show up anywhere on NBME material. But Q doesn't ride on you knowing it means hypokalemia to get it right. It's HY and pass-level to know that purging (anorexia or bulimia) causes hypokalemia anyway.
Delta wave	- Seen in Wolff-Parkinson-White syndrome (WPW; accessory conduction pathway in heart that bypasses the AV node, resulting in reentrant SVT). - Classically described as a "slurred upstroke" of the QRS, where the PR interval is shortened.

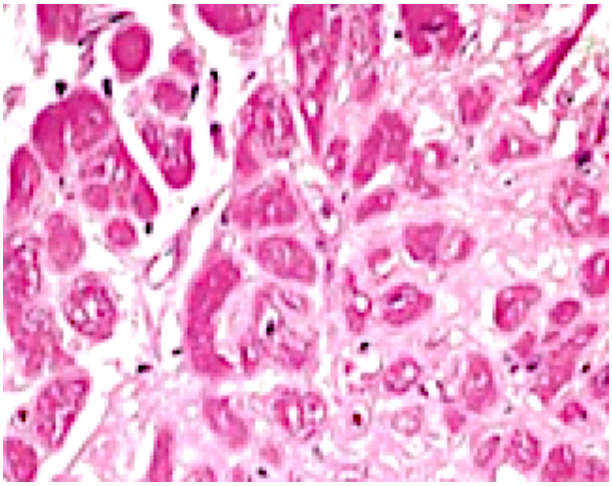
	 - Both the delta-wave and WPW have basically nonexistent yieldness on USMLE, but I mention them here so you are minimally aware.
J waves	- They mean hypothermia. You don't need to be able to identify on ECG. Just know they exist, as they show up in a 2CK vignette where patient has body temperature of 89.6 F (not 98.6).

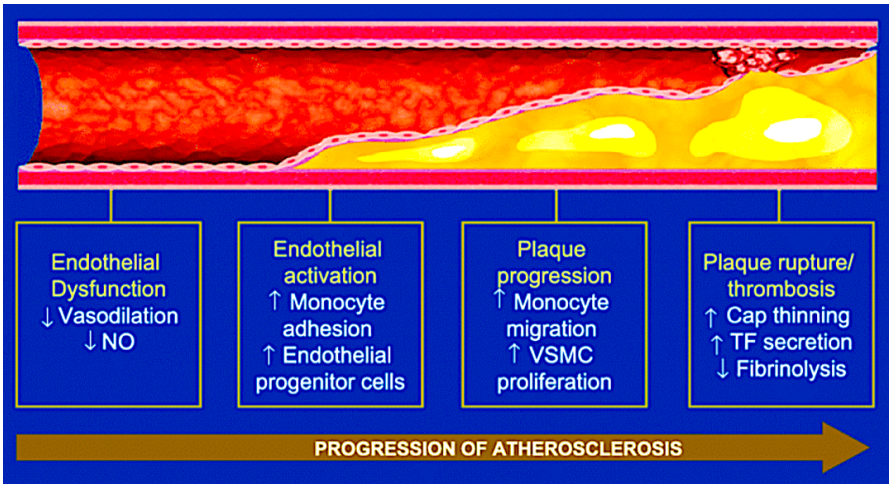
HY Heart block points for USMLE	
First degree	- Prolonged PR interval (>200 ms). Should normally be 80-120 ms.  - Note that above on the ECG, the PR-segment in particular (just prior to the QRS complex) is extra-long. - Not really assessed on USMLE. Just know the definition. - Don't treat on USMLE.
Second degree Mobitz type I (aka Wenckebach)	- Gradually prolonging PR interval until QRS drops. Then cycle repeats.  - Don't treat on USMLE.
Second degree Mobitz type II	- *No* gradual prolongation of PR interval, followed by a random dropping of the QRS.  - Can also sometimes occur as patterns of 2:1, 3:1, etc., where there will be a P to QRS ratio of 2:1 or 3:1, etc.  - Regardless as to whether the dropped QRS is random or in a numerical pattern, there is no gradual prolongation of the QRS before the dropped complex.

	- More dangerous than Mobitz I. This is because Mobitz II has higher chance of progression into type III heart block. - Treatment on USMLE is insertion of pacemaker. This is asked on a new 2CK NBME exam.
Third degree	- Two things you want to look for on ECG: - 1) Ultra-slow HR (i.e., 30-40). You'll see the QRS's are super far apart. This is the ventricular escape rhythm. - 2) No relationship between the P-waves and QRS complexes. Third degree AV block  - Treatment on USMLE is insertion of pacemaker. - So what you want to remember is that Mobitz II and 3rd-degree are the ones where we insert pacemaker; 1st-degree and Mobitz I we don't.

	HY Cardiomyopathy points for USMLE
Dilated (DCM)	- Can be isolated ventricular or diffuse 4-chamber dilation. Causes are multifarious, but a key feature is **systolic dysfunction,** where ejection fraction is reduced (i.e., <55%, where normal range is 55-70). - CXR shows enlarged cardiac silhouette.  - An S3 heart sound can sometimes be heard. - Cardiac exam shows lateralized apex beat. Can be described as the point of maximal impulse being in the anterior axillary line. Should be noted that this lateralization just means an enlarged LV, so it is non-specific, and can also be seen in LV hypertrophy from any cause. But many vignettes will mention it. - The arrows USMLE wants are: ↓ EF; ↑ LVEDV; ↑ LVEDP. I discuss this stuff in more detail in my HY Arrows PDF.

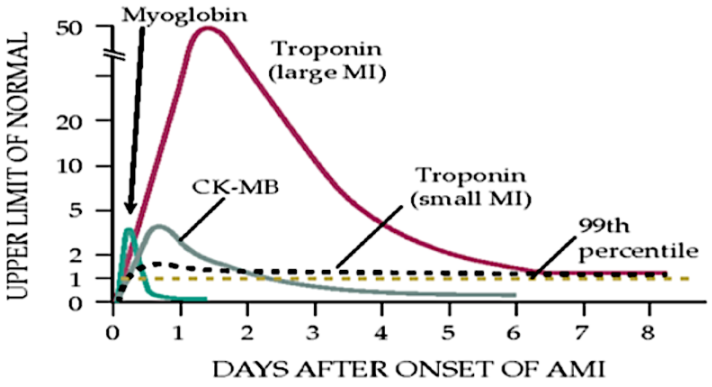
	- Causes of DCM are **ABCD**: - **A**: Alcohol. - **B**: Wet Beriberi (thiamine deficiency). - **C**: Coxsackie B virus, Cocaine, Chagas disease. - **D**: Drugs → doxorubicin (aka Adriamycin). - Other notable causes are: - Pregnancy (peripartum cardiomyopathy); hemochromatosis; rheumatic heart disease (myocarditis leading to DCM).
Hypertrophic (HCM)	- Heart failure due to systemic hypertension. - Characterized by diastolic dysfunction (the heart can pump just fine but cannot expand as easily). - Can be associated with S4 heart sound. - The arrows USMLE wants are: ↔ EF; ↔ LVEDV; ↑ LVEDP. - Ejection fraction is normal because the heart can pump perfectly fine. - Students get confused about LVEDV, thinking it should be low, if the heart cannot expand as easily. But this is not the case for USMLE. They want you to know normal volume can be achieved; *it just merely requires more force/pressure to get there.* - As with DCM, the apex beat / point of maximal impulse can be lateralized, which merely reflects LV hypertrophy. - As described earlier, paradoxical splitting of S2, left-axis deviation on ECG, and LBBB can all be seen due to LVH. When you see these findings in vignettes, don't get confused. They just mean LVH. - Often associated with hypertensive retinopathy (fundoscopy shows narrowing of retinal vessels, flame hemorrhages, and "AV-nicking"), hypertensive nephropathy (hyperplastic arteriolosclerosis with increased creatinine).
Hypertrophic obstructive (HOCM)	- Caused by mutations in β-myosin heavy-chain gene; autosomal dominant; results in disordered/disarrayed myocardial fibers. - HOCM causes asymmetric septal hypertrophy that results in the anterior *mitral* valve leaflet obstructing the LV outflow tract (so it can sound similar to aortic stenosis). - Classically sudden death in young athlete; cause of death is **ventricular fibrillation** due to acute left heart strain in the setting of fast heart rate. - Presents with systolic murmur that worsens with Valsalva or standing (aortic stenosis, in contrast, gets softer or experiences no change with Valsalva or standing). - HOCM and MVP are the only two murmurs that get worse with less volume in the heart. All other murmurs get worse with more volume. Valsalva increases intra-thoracic pressure and decreases venous return, so there's less volume in the heart. Standing simply decreases venous return. - NBME wants you to know that ↑ HR from exercise or stress means **"diastole is shortened more than systole."** - Beta-blockers (metoprolol or propranolol are both answers on NBME) are given to slow heart rate, which maximizes diastolic filling and decreasing symptoms / risk of death. -2CK: Implantable cardioverter-defibrillator (ICD) is indicated if patient develops any type of arrythmia, syncope, low blood pressure, or LV wall thickness >30mm. - Septal myectomy is done if the left ventricular outflow tract pressure gradient is >50mm Hg. But it should be noted ICD is usually first-line.
Restrictive (RCM)	- Heart failure due to diastolic dysfunction, where HTN is not the cause. - **JVD** is HY for RCM. An S4 can also be seen. The heart will not be dilated. - HY causes are Hx of radiation (leads to fibrosis), amyloidosis, and hemochromatosis.

	- Student might say, "I thought you said hemochromatosis was DCM. So if we have to choose on the exam, which one is it?" The answer is, whichever the vignette gives you. If they say a large cardiac silhouette with an S3 and lateralized apex beat, that's DCM. If they say JVD + S4 + nothing about a lateralized apex beat, you know it's RCM. - Amyloidosis is protein depositing where it shouldn't be depositing. Highest yield cause of amyloidosis on USMLE is multiple myeloma, which will lead to RCM.  Cardiac amyloidosis. Myocardium is pink; amyloid is white. - Since RCM is diastolic dysfunction, the arrows are the same as HCM, which are: ↔ EF; ↔ LVEDV; ↑ LVEDP.

Important points about atherosclerosis for USMLE	
Mechanism	- Endothelial cell damage / dysfunction (e.g., from HTN or smoking) → monocytes leave blood and accumulate beneath internal elastic lamina of artery → monocytes within tunica media of artery are now called macrophages → phagocytosis of oxidized LDL particles → macrophages are now called foam cells → accumulation of foam cells leads to fatty streak/plaque → stimulates vascular smooth muscle cell migration/proliferation within plaque → cap thinning + rupture → myocardial infarction, etc. 
HY points	- Most acceleratory risk factors are diabetes mellitus (I and II), followed by smoking, followed by HTN, in that order.

	- HTN is most common risk factor, but DM and smoking are worse. I talk a lot about this stuff in my HY Risk Factors PDF if you want extensive detail. - HTN is most acceleratory specifically for carotid stenosis (systolic impulse pounds carotids → endothelial damage). - Stroke, TIA, or retinal artery occlusion in patient with high BP is due to carotid plaque launching off to the brain/eye. If patient has normal BP, think AF instead, with left atrial mural thrombus launching off. - Patient over 50 with Hx of cardiovascular risk factors who now has accelerated HTN, think renal artery stenosis (narrowing due to atherosclerosis). - Plaques can calcify. The more calcium there is in a plaque, the more mature it is often considered to be. Calcium scoring is routinely done in patients who have coronary artery disease in the assessment of plaque progression. - Statins have 2 HY MOAs on USMLE: 1) inhibit HMG-CoA reductase; 2) upregulate LDL receptors on hepatocytes. - Ezetimibe blocks cholesterol absorption in the small bowel. - Bile acid sequestrants (e.g., cholestyramine) result in the liver pulling more cholesterol out of the blood. - Fibrates upregulate PPAR-α and lipoprotein lipase; best drugs to decrease triglycerides.

HY Angina points for USMLE	
Stable angina	- Chest pain that occurs predictably with exercise. - Due to atherosclerotic plaques causing >70% occlusion; can be calcific. - Classically causes ST depressions on ECG. - Nitrates (e.g., sublingual isosorbide dinitrate) used as Tx → nitrates "donate" nitric oxide (NO) that upregulates guanylyl cyclase within **venous** smooth muscle → increased cGMP → relaxation of venous smooth muscle → increased venous pooling of blood → decreased venous return → decreased myocardial oxygen demand → mitigation of chest pain. - Nitrates are contraindicated with PDE-5 inhibitors (e.g., Viagra) due to risk of low blood pressure. - Sodium nitroprusside used for hypertensive emergencies dilates **arterioles** in addition to the veins. If USMLE asks you where this drug acts, choose arterioles.
Unstable angina	- Chest pain that is unpredictable and can occur at rest. - Due to partial rupture of atherosclerotic plaque leading to partial occlusion. - ST depressions on ECG. - Diltiazem is answer on new 2CK NBME for patient with unstable angina. - Patients need cardiac catheterization.
Prinzmetal angina (variant angina pectoris)	- Vasospastic angina that occurs at rest (i.e., watching TV or while sleeping) in younger adults; it is not caused by atherosclerosis. - ST elevations are seen on ECG. - You must know that Prinzmetal is also known as variant angina pectoris. There is an NBME Q that gives vignette of Prinzmetal, but answer is "variant angina pectoris." - Treatment is nitrates (can cause coronary artery dilation unrelated to the venous pooling effects) or dihydropyridine calcium channel blockers (e.g., nifedipine). Avoid α1-agonists in these patients (cause vasoconstriction), as well as non-selective β-blockers like propranolol (can cause unopposed α effects).

HY histopath / cardiac marker points for MI	
Histopath	- Coagulative necrosis occurs within the first day of the MI. USMLE wants you to know cellular architecture is maintained in coagulative necrosis. - Neutrophils will infiltrate at 1-3 days. - Macrophages will infiltrate at 3-14 days. - Scar formation occurs from ~2 weeks onward.
HY random points	- Percutaneous coronary intervention used to open blocked coronary artery can cause *reperfusion injury*, where fast influx of oxygen to tissue results in high density of oxygen free radicals and peroxidation of lipid membranes → further myocardial necrosis. But this outcome is superior to non-intervention. - Adenosine is molecule that causes cardiac pain in MI and angina. It is main autoregulator of coronary blood flow. It is increased in MI due to exhaustion of ATP.
Cardiac markers	- USMLE wants you to know during acute MI, cardiac markers are not initially elevated. We rely on the ECG to show us ST elevations. - Troponin rises the most and stays elevated the longest. - CK-MB is the answer if they ask about recurrent MI. This is because its shorter duration of elevation makes it useful to discern a new cardiac event has occurred. 

Hypertensive Emergency + urgency (Steps 2CK/3 only)	
Emergency	- HTN >180/120 + signs of end-organ damage. - The latter can be hypertensive encephalopathy (confusion), nephropathy (poor renal function tests), retinopathy, acute heart failure, etc. - BP should be ↓ by no more than 20-25% in the first hour, as drastic ↓ can compromise perfusion to the brain and vital organs. - Blood pressure should be brought under 160/100 by 24-48 hours. - Drugs used are IV sodium nitroprusside, IV nicardipine, IV labetalol, and oral captopril.
Urgency	- HTN >180/120 + no signs of end-organ damage. - Blood pressure should be brought under 160/100 by 24-48 hours. - Drugs used are IV sodium nitroprusside, IV nicardipine, IV labetalol, and oral captopril.

HY Endocarditis points	
Acute endocarditis	- Bacterial infection of valve in patient with no previous heart valve problem. - Caused by *Staph aureus* on USMLE. - Left-sided valves (i.e., aortic and mitral) most commonly affected because of greater pressure changes (i.e., from high to low) within left heart, resulting in turbulence that enables seeding. - IV drug users → venous blood inoculated with *S. aureus* → travels to heart and causes vegetation of tricuspid valve.

Subacute endocarditis	- Bacterial infection of valve in patient *with history* of valve abnormality (i.e., congenital bicuspid aortic valve, Hx of rheumatic heart disease). - Caused by *Strep viridans* on USMLE. You need to know *S. viridans* can be further broken down into: *S. sanguinis, S. mutans,* and *S. mitis*. - Hx of dental procedure is HY precipitating event, where inoculation of blood occurs via oral cavity → previously abnormal valve gets seeded.
Random points	- New-onset murmur + fever = endocarditis till proven otherwise on USMLE. - Reactive thrombocytosis (i.e., high platelets) can occur due to infection. This is not unique to endocarditis, but it is to my observation USMLE likes endocarditis as a notable etiology for it. In other words, if you get an endocarditis question and you're like, "Why the fuck are platelets 900,000?" (NR 150-450,000), don't be confused. - Hematuria can occur from vegetations that launch off to the kidney. - Endocarditis + stroke-like episode (i.e., focal neurologic signs) = septic embolus, where a vegetation has launched off to the brain. - Janeway lesions, Osler nodes, splinter hemorrhages, etc., are low-yield for USMLE and mainly just school of medicine talking points. - HACEK organisms nonexistent on USMLE.
Management	- Blood cultures *before* antibiotics is important for 2CK. - Transesophageal echocardiography (TEE) confirms diagnosis after blood cultures. Transthoracic echocardiography (TTE) is not done for endocarditis. - For 2CK, empiric treatment for endocarditis is vancomycin, PLUS either gentamicin or ampicillin/sulbactam. - Vancomycin targets gram-positives (including MRSA). Gentamicin targets gram-negatives. - Endocarditis prophylaxis given prior to a dental procedure is usually ampicillin or a second-generation cephalosporin, such as cefoxitin. - Indications for endocarditis prophylaxis are: 1) Hx of endocarditis (obvious); 2) If there is any prosthetic material in the heart whatsoever; 3) If there is any congenital cyanotic heart disease that has not been completely repaired (if it's been completely repaired with prosthetics, give prophylaxis); 4) Hx of heart transplant with valvular regurgitation of any kind. - Highest yield point for USMLE about endocarditis prophylaxis is that mitral valve prolapse (MVP) and valve regurgitations or stenoses are ***not*** an indication. In other words, do not give prophylaxis if the patient has MVP, MR, AS, etc. In addition, bicuspid aortic valve is ***not*** an indication.

Rheumatic heart disease (rheumatic fever) HY points
- *Strep pyogenes* (Group A Strep) oropharyngeal infection results in production of antibodies against *S. pyogenes*' M-protein that cross-react with the mitral valve (i.e., molecular mimicry; type II hypersensitivity). - Can occur with the aortic valve in theory, but on USMLE, it is always mitral valve. - Results in mitral regurgitation acutely and mitral stenosis late, as discussed earlier. - Presents as **JONES** (J♥NES) → **J**oints (polyarthritis), **♥ C**arditis, subcutaneous **N**odules, **E**rythema marginatum (annular, serpent-like rash), **S**ydenham chorea (autoimmune basal ganglia dysfunction that results in dance-like movements of the limbs). - Cutaneous Group A Strep infections don't cause rheumatic fever, but can still cause PSGN. - Treatment is penicillin.

Conditions confused for cardiac path	
Panic attack	- Psych forms love trying to make you think this is an MI. - They'll give you young, healthy patient who feels doom / like he or she is going to die. - Sometimes they mention in stem Hx of MI in family as distraction. - They can say patient has mid-systolic click, as discussed earlier, and then they ask for cause of patient's symptoms → answer = panic disorder, not MVP. Student gets confused, but MVP is almost always asymptomatic, where panic attack is clearly cause of the patient effusively hyperventilating. - Treat with benzo.
Orthostasis	- Orthostatic hypotension is defined as intravascular fluid depletion causing a drop of systolic BP >20 mmHg and diastolic BP >10 mmHg when going from supine to standing. - Shows up on 2CK IM form as exactly a drop of 20 and 10, respectively, for systolic and diastolic BPs in a patient with fainting → answer = "intravascular fluid depletion." - Diuretic use is big risk factor.
Vasovagal syncope	- Fainting in response to stressor (e.g., emotional trigger). - Stress triggers an initial sympathetic response, which in turn triggers a compensatory parasympathetic response. This latter response is excessive in some people, where the peripheral arterioles dilate and the heart slows too much → decreased cerebral perfusion → lightheadedness/fainting. - 2CK wants you to know a tilt-table test can be used to diagnose, where a reproduction of symptoms can occur.
Carotid sinus hypersensitivity	- USMLE likes this for both Steps 1 and 2. - They'll say dude was shaving then got lightheadedness or fainted. Mechanism is ↑ stretch of carotid sinus baroreceptors → ↑ afferent CN IX firing to solitary nucleus of the medulla → ↑ efferent CN X parasympathetic firing down to cardiac nodal tissue → ↓ HR → ↓ CO → ↓ cerebral perfusion.
Costochondritis	- Inflammation of cartilage at rib joints. - Will present as chest pain that worsens with palpation or when patient reaches over the head or behind the back. These two findings are clear indicators we have an MSK condition, not cardiac. - Can be idiopathic, caused by strain (e.g., at the gym), or even post-viral.
Pleurodynia	- MSK condition asked twice on NBMEs that has nothing to do with the lungs, despite the name. - This is viral infection (Coxsackie B) causing sharp lateral chest pain due to intercostal muscle spasm. Sometimes students choose pericarditis, etc., even though the presentations are completely disparate. - Creatine kinase can be elevated in stem due to ↑ tone of muscle.
Viral pleurisy	- Viral infection causing inflammation of the pleura (layers covering the lungs), leading to sharp chest pain. - If this is the answer, CK will be normal (unlike pleurodynia, because it's not MSK).
Diffuse esophageal spasm	- Can cause angina-like pain in patient without cardiovascular disease. - I discuss this in detail in HY Gastro PDF.
Gastroesophageal reflux	- Can present as chest pain confused for MI. ECG will be normal, clearly. - I discuss GERD in detail in HY Gastro PDF.

<table>
<tr><th colspan="2">Arterial vs venous disease</th></tr>
<tr><td>Arterial disease</td><td>- Caused by atherosclerotic disease; presents as diminished peripheral pulses in patient over 50 who has risk factors – i.e., diabetes, smoking, HTN.
- Lower legs can be shiny and glabrous (trophic changes).
- Arterial ulcers are small and punched-out; located on tops/bottoms of feet and toes.
 
- Ankle-brachial indices (ABIs) are first step in diagnosis (exceedingly HY on 2CK), which compare BP in ankle to the arm; if <0.9, this reflects ↓ peripheral blood flow due to atherosclerosis.
- If ABIs are not listed as first step in diagnosis for whatever reason, choose Doppler ultrasound. There is one 2CK NBME Q where this is the case.
- After ABIs, next step is exercise stress test (if listed) in order to determine exercise tolerance. If not listed, go straight to “recommend an exercise / walking program.” Do not choose cilostazol first or arteriography as answers.
- CMS Surg form 5 has “prescription for an exercise program” as the answer. Students say, “Why does it say ‘prescription’?” No fucking idea. Ok?
- 2CK only: Surgery is indicated in the event of critical limb ischemia, which is when there is chronic ischemic rest pain, ulcers, or gangrene.
- All patients with arterial disease should be on triad of 1) ACEi/ARB, 2) statin, 3) anti-platelet therapy (same as carotid stenosis). This is unrelated to the management sequence of exercise program → cilostazol → surgery.</td></tr>
<tr><td>Venous disease</td><td>- Congestion of venous system usually from valvular incompetence; idiopathic / familial; varicose veins are one type of venous disease and are not synonymous; patients can have venous disease without varicosities.
- Peripheral pulses are normal (those reflect arteries, not veins).
- Lower legs demonstrate “brawny edema,” which is a brown, hemosiderin-laden edema due to ↑ pressure / micro-extravasations; hyperpigmentation resulting in brown/red skin is known as stasis dermatitis, aka post-phlebitic syndrome; the latter is a term is asked on 2CK, so know the annoying vocab.
- Venous ulcers are large and sloughy, and located at the malleoli.
 
- Diagnose with venous duplex ultrasonography of the legs; first treatment is compression stockings. Never choose answers such as venous stripping or glue agents, etc.</td></tr>
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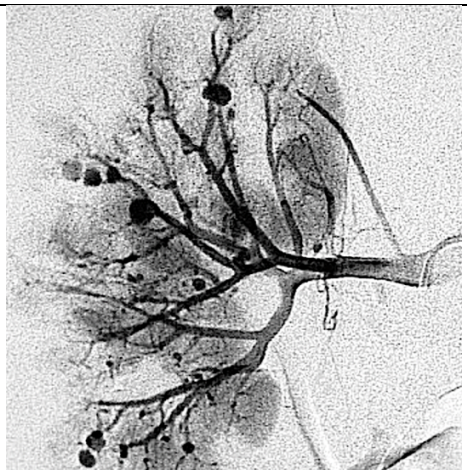
	- Venous disease ↑ risk for DVT and superficial thrombophlebitis. If patient has active DVT or STP, answer = subcutaneous enoxaparin (heparin) over compression stockings. - 2CK: Surgery is indicated if there are significant skin changes, venous ulcers, or for symptomatic varicose veins (i.e., pain, itching, swelling, cramps). - You could be aware that sclerotherapy is often chosen as the first surgery, where a sclerosing (scarring) agent is injected into the vein, causing it to close where the blood is re-routed to other veins. But basically always, just compression stockings, or rarely heparin for DVT or STP, is the answer.

Cardiac stress test points (2CK only)	
- Most 2CK Qs that ask about stress tests are in the context of evaluating patients for perioperative MI risk. - It is rare the Q will force you to choose between different types of stress tests. 4/5 Qs will just list one stress test, where it is simply assessing, "Do you know a stress test should be done, period, in this scenario." - Stress tests are also done for peripheral arterial disease prior to recommending an exercise/walking program (as mentioned above).	
Exercise ECG	- Most common stress test. - The answer on USMLE for patients who have stable angina, where you're looking for ST depressions (i.e., evidence of ischemia) with exertion. - **Requires a patient has a normal baseline ECG in order to perform.** - In other words, the Q will give you a big 15-line paragraph + mention in the last line that the patient's baseline ECG shows, e.g., a LBBB from a year ago that's unchanged. This means ECG stress test is wrong in this situation, since you need to have a normal ECG to do it. The 1/5 Qs that force you to choose between stress tests want you to know this detail, basically, where you just choose the non-ECG stress test instead.
Exercise echo	- Used to look for heart failure (i.e., ↓ EF) with exertion, not overt ischemia. - In other words, the answer on USMLE for patients who don't get chest pain with exertion (i.e., don't have stable angina), but who get shortness of breath with exertion. This reflects, at a minimum, left heart decompensation with possible ↓ EF. - Also the answer for patients who have abnormal baseline ECG.
Pharmacologic	- Refers to numerous answer choices on USMLE – i.e., dobutamine-echo, dipyridamole-thallium. - The answer on USMLE for patients who cannot exercise, such as in the setting of angina when merely walking up a single flight of steps, or in patients imminently undergoing major surgery (e.g., AAA repair), where perioperative MI risk needs to be assessed. I have seen both of these scenarios on 2CK forms. - The USMLE will typically not force you to choose between stress tests. As I mentioned at the top of this table, they will usually just have the pharmacologic stress test as the only one listed. - Dobutamine is a β1-agonist that stimulates the heart (i.e., ↑ oxygen demand). Echo can then be done to look for ↓ EF (i.e., heart failure). - Dipyridamole is a phosphodiesterase inhibitor that dilates arterioles. HR goes up to compensate, thereby ↑ myocardial oxygen demand. Thallium is then used to look at perfusion of the myocardium.
Myocardial perfusion scan	- "Myocardial perfusion scan" refers to using any radiotracer (i.e., thallium, technetium, sestamibi) to evaluate the heart as part of a stress test. - Whereas pharmacologic stress test uses a radiotracer only when the patient is not exercising, a myocardial perfusion scan is a broader term that can refer to using a radiotracer during exercise also. - Examples are exercise-thallium and exercise-sestamibi stress tests. - The latter two, however, are not called pharmacologic stress tests (PST) because they involve exercise. So a PST is a type of MPS, but a MPS is not a type of PST.
Cardiac scintigraphy	- "Cardiac scintigraphy" is a broad term that refers to any evaluation of the heart in which some form of radiotracer is used (i.e., thallium, technetium, sestamibi), even if not as part of a stress test.

	- In other words, cardiac scintigraphy can be done to merely investigate cardiac anatomy or inflammatory processes unrelated to the evaluation of ischemia or heart failure. - On USMLE, however, it is to my observation that cardiac scintigraphy, myocardial perfusion scan, and pharmacologic stress test are all used interchangeably. - So you might get a Q where a patient can't exercise due to immediate chest pain, and the answer is just "cardiac scintigraphy," or "dipyridamole-echocardiography stress test."

Knee trauma causing popliteal arterial injury	
Knee dislocation	- If Q gives you MVA where the knee is injured + absent pulses distally, the sequence of answers they want is knee relocation first, followed by arteriography to look for popliteal artery injury. - In one of the 2CK Qs, they already tell you the knee is relocated, then the answer is "arteriography with runoff." Students say, "what's the runoff part?" No fucking idea, it's just what they want.
Penetrating trauma	- If Q gives gunshot wound to the knee + absent distal pulses, go straight to "surgical exploration" as the answer. - This could be thought of as the knee-equivalent of a gunshot wound to the abdomen, where straight to laparotomy (even if patient is stable) is the answer.

HY Vasculitides (fancy word that's pleural for vasculitis)	
Granulomatosis with polyangiitis	- Formerly known as Wegener granulomatosis. - Answer on USMLE for adult with triad of 1) hematuria, 2) hemoptysis, and 3) "head-itis" – i.e., any problem with the head, such as nasal septal perforation, mastoiditis, sinusitis, otitis. - Associated with cANCA and anti-proteinase 3 (anti-PR3) antibodies. - Causes "necrotizing glomerulonephritis" that can lead to rapidly progressive glomerulonephritis (RPGN).
Eosinophilic granulomatosis with polyangiitis	- Formerly known as Churg-Strauss. - Presents as combo of asthma + eosinophilia +/- head-itis. - Head-itis always seen in Wegener vignettes, but maybe only ~50% of CS Qs. - Renal involvement rare for CS. - Associated with pANCA and anti-myeloperoxidase (anti-MPO) antibodies.
Microscopic polyangiitis	- Will just present as hematuria in a patient who is pANCA / anti-MPO (+). - Similar to Wegener, can cause RPGN.
- Vignette on USMLE can give you Wegener, Churg-Strauss, or microscopic polyangiitis, and then the answer is just "vasculitis," or "polyangiitis." - The above three conditions can be associated with a weird neuropathy called mononeuritis multiplex, which means neuropathy of "one large nerve in many locations" – e.g., wrist drop + foot drop in same patient. - You don't need to worry about the fancy term "mononeuritis multiplex," but what I do want you to be aware of is that neuropathy will sometimes show up in these vasculitis vignettes, so don't be confused about it.	
Polyarteritis nodosa (PAN)	- Medium-vessel vasculitis that causes a "string of pearls" appearance of the renal vessels.

	 - Causes fibrinoid necrosis, which means it looks like fibrin but it ain't fibrin. - Offline Step 1 NBME has "segmental ischemic necrosis" as the answer. - **Can be caused by hepatitis B.** - For whatever reason, USMLE wants you to know PAN spares the lungs – i.e., it does not affect the pulmonary vessels.
Takayasu arteritis	- Aka "pulseless disease." Classically affects Asian women 40s or younger. - Inflammation of large vessels, including the aorta. - Always affects the subclavian arteries (which supply the arms), which is why it can cause weakly, or non-palpable, pulse in the upper extremities.
Temporal arteritis	- Aka giant cell arteritis. - 9/10 Qs will be painful unilateral headache in patient over 50. I've seen one Q on NBME where it's bilateral. - Flares can be associated with low-grade fever and high ESR. - Patients can get proximal muscle pain and stiffness. This is polymyalgia rheumatica (PMR). The two do not always go together, but the association is HY. (Do not confuse PMR with polymyositis. The latter will present with ↑ CK and/or proximal muscle weakness *on physical exam*. PMR won't have either of these findings. I talk about this stuff in detail my MSK notes.) - Patients can get pain with chewing. This is jaw claudication (pain with chewing). - Highest yield point is we give **steroids before biopsy** in order to prevent blindness. - An NBME has "ischemic optic neuropathy" as the answer for what complication we're trying to prevent by giving steroids in temporal arteritis. - IV methylprednisolone is typically the steroid given, since it's faster than oral prednisone. - It's to my observation many 2CK NBME Qs will give the answer as something like, "Steroids now and then biopsy within 3 days," or "IV methylprednisolone and biopsy within a week." Students ask about the time frames, but for whatever reason USMLE will give scattered/varied answers like that. - Another 2CK Neuro CMS Q gives easy vignette of temporal arteritis and then asks next best step in diagnosis → answer = biopsy. Steroids aren't part of the answer. Makes sense, since they're asking for a diagnostic step.
Thromboangiitis obliterans	- Aka Buerger disease; technically a vasculitis. - Dry gangrene of the fingers or toes seen generally in male over 30 who's a heavy smoker. - Treatment is **smoking cessation.** - Don't confuse with Berger disease, which is IgA nephropathy.
Ascending aortitis	- Tertiary syphilis can cause ascending aortitis + aortic aneurysm.

	- *T. pallidum* invades vasa vasorum (small vessels supplying the wall of the aorta itself) causing endarteritis (inflammation of inner lining of the aorta) and obliteration of the vasa vasorum. Can cause "tree-barking" of the aorta.

Thrombophlebitis	
- Phlebitis = inflammation of a vein. - Thrombophlebitis = inflammation of a vein due to a thrombus.	
Deep vein thrombosis	- DVT will be unilateral thigh or lower leg swelling in patient with risk factors such as: post-surgery, prolonged sedentation, OCP use, Hx of thrombotic disorders (e.g., Factor V Leiden, prothrombin mutation). - Virchow triad for ↑ DVT risk: 1) venous stasis (e.g., post-surgery sedentation), 2) hypercoagulable state (e.g., estrogen use, underlying malignancy), 3) endothelial damage (i.e., smoking). - Combined OCPs contraindicated in smokers over 35 because estrogen causes hypercoagulable state for two reasons: 1) estrogen upregulates fibrinogen; 2) estrogen upregulates factors Va and VIIIa. - USMLE loves nephrotic syndrome as cause of DVT (loss of antithrombin III in the urine → hypercoagulable state). - Antiphospholipid syndrome → DVTs despite paradoxical ↑ PTT (i.e., if PTT is high, you'd think you have bleeding diathesis, not thromboses); may or may not be due to SLE. Antibodies against phospholipids cause *in vivo* clumping of platelets + ↑ clot initiation, but disruption of *in vitro* PTT assay means ↑ PTT. - Major danger is DVT can embolize to lungs causing PE → acute-onset shortness of breath and tachycardia + death if saddle embolus. - Homan sign can mean DVT, which is pain in the calf with dorsiflexion of foot. - Diagnose DVT with duplex venous ultrasound of the leg/calf. - Treatment is heparin. - Harder surgery stuff for 2CK is that they care about prophylactic vs therapeutic doses of heparin. Prophylactic dose is lower-dose and is used perioperatively in patients with venous disease/stasis or who are high risk. If a patient has an actual full-blown DVT, however, give therapeutic dose, which is higher-dose. - There are two 2CK Qs on this stuff. One just mentions a guy going into surgery who has Hx of venous stasis → answer = "prophylactic heparin dose"; "therapeutic heparin dose" is wrong answer. - The second question gives a guy *who's already on prophylactic heparin* but gets a DVT anyway. The answer is then "heparin." It's weird because students are like, "Wait what? He's already on heparin though." And I'm like, yeah, but what they mean is, we have to give therapeutic dose now for the active DVT, which is higher dose. - DVT can rarely cause stroke if an ASD is present (paradoxical embolus). Dumb and low-yield, but it shows up, and students get fanatical over it.
Post-op migratory	- Post-surgery, this is usually due to changes in hemostasis and coagulability. - Will present as pink/red painful lesions appearing asymmetrically on the limbs within days of surgery. You just need to be able to diagnose this.
Trousseau sign of malignancy	- Migratory thrombophlebitis classically due to head of pancreas adenocarcinoma. But this can also be seen with adenocarcinomas in general, e.g., pulmonary. - One proposed mechanism is liberation/secretion of tissue factor (factor III) by adenocarcinomas, thereby activating the clotting cascade. - Don't confuse with Troisier sign of malignancy, which is a palpable left superclavicular lymph node (Virchow node), often due to gastric or other visceral cancer.
Catheter-associated septic thrombophlebitis (CAST)	- Shows up on 2CK Surg form as patient who had a catheter in and then develops a 4-cm indurated, painful, fluctuant cord in his arm (refers to vein).

	- Answer = "excision of vein." Obscure question, but not my opinion. Take it up with NBME if you think it's weird.
Pelvic septic thrombophlebitis (PST)	- Particularly HY on 2CK Obgyn forms. - The answer on USMLE in a woman who has post-partum endometritis (fever + tender lower abdomen) with persistent fever >48 hours despite antibiotics. - Endometritis can lead to ↑ risk of local infective clots in the ovarian veins. - If they give you a post-partum woman with sepsis (i.e., SIRS + infection), but the vignette doesn't fit PST as described above, the answer is "puerperal sepsis" on the Obgyn form. The latter is a more general term and can refer to many causes of post-partum sepsis (including PST confusingly enough).
Superficial thrombophlebitis	- Painful palpable cord in the ankle that may or may not track up to the knee. - Seen in patients with venous insufficiency. - Answer is "subcutaneous enoxaparin." Compression stockings are typically the answer for first step in venous insufficiency, but if you have an active ST or DVT, heparin must be given as first step.

HY familial dyslipidemias for IM		
Condition	**Mechanism**	**HY point(s)**
Hyperchylomicronemia (AR)	Deficiency of lipoprotein lipase or Apolipoprotein C-II	- ↑ chylomicrons + TGAs. - Pancreatitis (abdo, not chest, pain). - Xanthomas. - Plasma appears "creamy".
Hypercholesterolemia (AD)	Deficiency (heterozygous) or absence (homozygous) of LDL receptor or Apo B-100	- ↑ LDL. - LDL usually 3-400 in heterozygotes, with MI/death in 30-40s (answer = deficiency of functional LDL receptor). - LDL usually 700-1000 in homozygotes, with MI in teens (answer = absence of functional LDL receptor). - Xanthomas.
Hypertriglyceridemia (AD)	↑ Hepatic production of VLDL	- ↑ TGAs. - Pancreatitis.

Random HY cardio pharm points	
Verapamil	- Non-dihydropyridine calcium channel blocker (acts on nodal calcium channels). - **Causes constipation** (HY on Family Med). - Used for AF for rate control sometimes in place of metoprolol (don't worry about the use-case; USMLE doesn't give a fuck; just know MOA and side-effect).
Nifedipine	- Dihydropyridine calcium channel blocker (acts on arteriolar calcium channels → dilates arterioles → ↓ peripheral vascular resistance → ↓ BP). - **Causes peripheral edema / fluid retention** (HY on Family Med). - Used for essential HTN in patients without diabetes, atherosclerotic disease, or renal disease (if that sounds confusing, I talk about this in extensive detail in my HY Risk Factors PDF; this is strict 2CK FM talking point).
β-blockers	- Beta-1 is on myocardium. - Beta-2 is on peripheral arterioles. - There are other binding spots, but the above are what USMLE cares about 99%. - Beta-1-selective blockers follow the mnemonic: **A BEAM** of beta-blockers: **A**tenolol, **B**isoprolol, **E**smolol, **A**cebutolol, **M**etoprolol. - The important beta-1/2-non-selective blocker is propranolol. - Labetalol and carvedilol antagonize both beta-1 and alpha-1. They don't end in –olol, which makes them easier to remember/group this way.

	- Beta-1 antagonism slows HR → ↑ diastolic filling time → ↑ LVEDV. Even though ↑ preload ↑ oxygen demand, the ↓ chronotropy and inotropy cause a ↓↓ in oxygen demand, where the net effect is ↓ oxygen demand. - Step 1 NBME wants beta-blockers as "slowing the rate of diastolic depolarization." In other words, if HR slows, then fractionally more time is spent in diastole, which means the process of returning to systole is delayed/protracted. - Labetalol used first-line in patients who have aortic dissection and traumatic rupture of the aorta. Nitroprusside comes after. - 2CK Q gives "esmolol + nitroprusside" as answer to a traumatic rupture Q, but almost always, they will just want "labetalol." - Timolol can be used topically for glaucoma (↓ aqueous humor production). - Propranolol used for innumerable things → Tx of tachycardia in hyperthyroidism (because also has additional effect of ↓ peripheral conversion of T4 → T3); akathisia (restlessness as an extra-pyramidal side-effect of antipsychotics); essential, alcoholic, or idiopathic tremors (have seen various tremors treated on NBME forms this way, not just essential tremor); social phobia (if patient has asthma, 2CK Psych form wants benzo instead); HOCM (↓ HR → ↑ LVEDV → ↓ murmur/obstruction). - For propranolol, USMLE wants: ↑ LVEDV; ↓ CO, ↑ peripheral vascular resistance. - Propranolol is β1/2-non-selective; β2 agonism normally has dilatory effect on peripheral arterioles, so if we antagonize → ↑ peripheral vascular resistance. - For labetalol, USMLE wants: ↑ LVEDV; ↓ CO, ↓ peripheral vascular resistance. - Labetalol, in contrast, has some α1 blockade effect in addition to β1, so PVR is ↓. - Beta-blocker given after ACEi/ARB in Tx of heart failure (i.e., choose ACEi or ARB first). - Metoprolol used for rate-control in AF. - Beta-blockers can cause depression and sexual dysfunction. - Avoid in patients with lung disease or history of severe or psychotic depression.
Dobutamine	- Beta-1 agonist used in pharmacologic stress testing (i.e., dobutamine-echo).
Amiodarone	- Potassium channel blocker. - Can cause TdP, greyish skin discoloration, and thyroiditis. - Used for VT in patients without coma or low BP.
Quinidine	- Sodium channel blocker. - Can cause TdP and cinchonism (headache + tinnitus).
Digoxin	- Directly blocks myocardial Na^+/K^+ ATPase pump → causes indirect inactivation of myocardial Na^+/Ca^{2+} ATPase → more Ca^{2+} remains in myocardial cell → increased contractility. - Also has parasympathomimetic effect at nodal tissue that slows HR. - In other words, digoxin both slows HR + increases contractility. - Hypokalemia can cause toxicity. This is because digoxin binds to extracellular K^+ binding site, so if less K^+ is around to compete, lower dose is needed to induce effect. - Toxicity presents classically as yellow/wavy "Vincent van Gogh" vision. - Does not decrease mortality in heart failure.
Furosemide	- Classic loop diuretic that inhibits the apical $Na^+/K^+/2Cl^-$ ATPase symporter on the thick ascending limb. Apical means side of urine. Basolateral means side of blood. - Most efficacious diuretic at enabling fluid unloading for pulmonary and peripheral edema. Do not decrease mortality in cardiac patients. - Answer on USMLE for patient with heart failure who has ↓ O2 sats who "refuses to lie supine/back on the gurney" because he/she can't breathe. Q might say CXR or exam shows fluid 2/3 up the lung fields. - Application of loops can be confused by students with ACEi/ARB. For instance, the latter are used first-line to ↑ EF in heart failure even if O2 sats are low (↓ AT-II constrictive effect on arterioles → ↓ afterload → heart pumps easier), but loops are first-line if the vignette *specifically emphasizes* the dyspnea and low O2 sats of the patient as the focus. - HY point is that they can cause **hypokalemia.** What USMLE loves to do is tell you a patient is initiated on furosemide but it is insufficient + now needs a second diuretic →

	answer = anything that is potassium-sparing (i.e., ENaC inhibitor such as amiloride or triamterene; or aldosterone receptor antagonist such as spironolactone or eplerenone). - Can also cause ototoxicity (tinnitus, vertigo). - Loops ↑ urinary calcium. This is in contrast to thiazides, which ↓ it. - Ethacrynic acid is another loop you can be aware of. It is used in patients with sulfa allergies, since furosemide is a sulfa drug. It is also ototoxic.
Thiazides	- Hydrochlorothiazide (HCTZ) is HY example; chlorthalidone is "thiazide-like." - Inhibit the Na^+/Cl^- symporter on the apical membrane of the early-DCT. - ↑ calcium reabsorption in the DCT of the kidney. Used to ↓ recurrence of calcium stones by promoting ↑ reabsorption of urinary calcium. In turn, they can sometimes cause hypercalcemia. There is an NBME Q where they ask for MOA of HCTZ, where Na^+/Cl^- symporter inhibition isn't listed; answer is just "↑ calcium reabsorption." - Thiazides and dihydropyridine calcium channel blockers are used first-line for HTN in patients without any renal or cardiovascular issues. If patient has proteinuria, ↑ in creatinine or renin, or pre-diabetes or diabetes, ACEi or ARBs are used first. - Can be used in heart failure in patients with diabetic nephropathy who are already on ACEi and β-blocker. Sounds specific, but there's a new 2CK Q that has thiazide as correct over spironolactone in a diabetic. Spironolactone can cause hyperkalemia in patients who have worrisome kidney function. - Thiazides can cause gout (i.e., contraindicated in gout). - Offline NBME 20 wants you to know thiazides can cause galactorrhea (milky discharge from the nipples) via some obscure mechanism. Literature search shows it accounts for ~0.2% of adverse effects of thiazides, but count on NBME to ask it.
Amiloride, Triamterene	- ENaC inhibitors → block apical sodium channel in cortical collecting duct → ↓ Na^+ reabsorption → ↓ water reabsorption. - These are potassium-sparing, which means they do not ↓ serum K^+. This is because by inhibiting the apical ENaC channel, they indirectly inhibit the basolateral Na^+/K^+ ATPase → ↓ Na^+ reabsorption + ↓ K^+ secretion. - The answer on USMLE for a second diuretic given in a patient already on furosemide who needs additional fluid unloading. Be careful however. I've seen one Q where the patient is on furosemide, but the point of the Q is he/she needs HTN control, and the answer is a thiazide, not the ENaC inhibitor. You want to select an ENaC inhibitor specifically if the Q says, "We have a patient who has peripheral/pulmonary edema due to X cause + is already on furosemide; what do we do now?" → answer = amiloride or triamterene.
Spironolactone, Eplerenone	- Aldosterone receptor antagonists. - By blocking aldosterone receptor, they ↓ activity of the basolateral Na^+/K^+ ATPase → leads to indirect ↓ activity of apical ENaC → ↓ Na^+ reabsorption → ↓ water reabsorption. - Potassium-sparing. Same as with ENaC inhibitors, used for fluid unloading in patients who are already on furosemide in whom we worry about dropping their K^+ too much. - Used in heart failure up the hierarchy of meds – i.e., ACEi/ARB first-line, followed by adding β-blocker, followed by spironolactone (but if patient is diabetic with poor renal function, don't add spironolactone here; give thiazide instead as per new 2CK NBME). - Can be for aldosteronoma (Conn syndrome) prior to surgery. - Spironolactone can cause **gynecomastia** (anti-androgenic effect by blocking androgen receptors). - Eplerenone has ↓ risk of gynecomastia compared to spironolactone.
Lisinopril	- USMLE-favorite ACE inhibitor. - Prevents conversion of AT-I into AT-II in the lungs. - Used for HTN in patients with pre-diabetes, diabetes, atherosclerotic disease, or renal disease (I talk about this in HY Risk Factors PDF in more detail). - Can cause dry cough (ACE is aka bradykininase, so ACEi cause ↓ breakdown of bradykinin in lungs → cough). - Can ↑ serum K^+, since ↓ aldosterone synthesis. Aldosterone normally secretes K^+ in the distal kidney). - Avoid in hereditary angioedema.

Valsartan	- Angiotensin II receptor blocker (ARB). - Use-cases are identical on USMLE to ACEi (i.e., if you see both as answer choices to a question, they're usually both wrong because they're the "same"). - Doesn't cause dry cough the way ACEi do.
Oxymetazoline, Phenylephrine	- α1 agonists → constrict arterioles → ↑ BP → HR ↓ due to baroreceptor reflex. - Highest yield uses on USMLE are for **nasal decongestion** → constrict capillaries within nasal mucosa → ↓ inflammation → relief of congestion. - Can cause rhinitis medicamentosa, which means rebound nasal congestion upon withdrawal if used non-stop for ~5 days. In other words, patients should use only as needed for a maximum of about ~3-4 days while sick.
Isoproterenol	- β1/2 agonist → increases HR and decreases peripheral vascular resistance.
Methyldopa, Clonidine	- α2 agonists → prevent presynaptic release of neurotransmitters, especially NE and S. - Methyldopa used for HTN in pregnancy (nifedipine and labetalol also used). - Clonidine used for various psych treatments (e.g., Tourette).
Mirtazapine	- α2 antagonist. - Used to treat depression in patients who have anorexia (stimulates appetite).
Ritodrine	- β2 agonist used as tocolytic (i.e., slows/delays labor).
Hydralazine	- Dilates arterioles → ↓ BP → HR goes ↑ to compensate. - Used for hypertensive emergencies in pregnancy. - Affects calcium currents (but not a calcium channel blocker).
Nitrates	- As discussed earlier, they liberate NO which ↑ guanylyl cyclase → ↑ cGMP → relaxation of venous smooth muscle → venous dilation/pooling → ↓ preload on heart → ↓ oxygen demand → relief of anginal pain. - Don't combine with PDE-5 inhibitors (e.g., sildenafil), which prevent breakdown of cGMP. This can lead to severe hypotension. - Combination of hydralazine + nitrates decreases mortality in heart failure. - For sodium nitroprusside, choose arterioles as site of action.
Statins	- Statins have 2 HY MOAs on USMLE: 1) inhibit HMG-CoA reductase; 2) upregulate LDL receptors on hepatocytes. - Can cause myopathy and toxic hepatitis. An offline Step 1 NBME has myopathy as correct over toxic hepatitis. - Indications for statins on 2CK vary depending on the source (i.e., whether to use the 70 vs 100 mg/dL cutoff in certain scenarios), but for FM/USMLE, give if: - Age 20-39 if LDL > 190 mg/dL. - Age 40-75 if LDL > 100 mg/dL. - Age 20-75 in diabetic if LDL >100 mg/dL. - Some sources use 70, rather than 100, for the latter two cutoffs. What I can say is that I've seen 2CK NBME Qs where they use 100 as the cutoff (i.e., they give an LDL of 95 and statin is wrong, implying LDL is satisfactory). - Some sources incorporate a CVD risk % of >7.5%, which is more of a moot / pedantic talking point. I've seen one 2CK NBME Q where a CVD risk % shows up in the stem, but it doesn't rely on you knowing that point to get it right.
Fibrates	- Fibrates (e.g., fenofibrate) upregulate PPAR-α and lipoprotein lipase; best drugs to decrease triglycerides. - Give if triglycerides >300 mg/dL. - Can cause myopathy and hepatotoxicity, as well as cholesterol gall stones. - If the USMLE asks you why statins + fibrates combined have ↑ chance of myopathy, the answer is "P-450 interaction."
Ezetimibe	- Blocks cholesterol absorption in the small bowel, thereby ↓ LDL. - USMLE just wants you to know MOA. - Doesn't decrease mortality.
Cholestyramine	- Bile acid sequestrant. ↓ LDL. Doesn't ↓ mortality. - Causes reduced enterohepatic circulation of bile acids at terminal ileum → liver must now convert more cholesterol into bile acids in order to replenish them → liver pulls cholesterol out of the blood to accomplish this.
Orlistat	- Pancreatic lipase inhibitor used for obesity; can cause steatorrhea.

Niacin	- Vitamin B3. Two MOAs USMLE wants you to know: 1) ↓ VLDL export by the liver; 2) ↑ HDL more than any other medication. - Administration can cause flushing (caused by prostaglandin, so mitigated by asprin), gout, and insulin resistance. - Deficiency → pellagra → 3Ds → dementia, dermatitis, diarrhea. - Can present as delirium instead of overt dementia (biggest risk factor for delirium is underlying dementia, so old patient with delirium often has underlying cognitive decline); dermatitis will present on USMLE as either hyperpigmentation of the skin of the forearms or Casal necklace.
Procainamide	- Sodium channel blocker that can be used in Wolff-Parkinson-White. - Can cause drug-induced lupus with anti-histone antibodies.
Flecainide	- Na^+ channel blocker used as first-line rhythm control in patients fail rate-control for AF. - Patient should have no structural or coronary artery disease. Otherwise use a potassium channel blocker like amiodarone or dofetilide.
Dofetilide, Ibutilide	- Potassium channel blockers. - Can cause torsades (asked on NBME).
Omega/fish oils	- Decrease triglycerides.
Morphine	- 2CK Surgery form gives patient who is in pain from surgery despite being on max doses of morphine + they say blood pressure is elevated + ask what is next best step → answer = "increase bolus of morphine." Unusual, since they say max dose of morphine already, but 1) we always full treat pain, and 2) explanation for the Q talks about how pain can lead to high BP. - My point here is that just be aware HTN can be caused by pain → treating pain can bring down the BP. - Separately, treatment of MI, acute limb ischemia, and sickle crisis usually includes oxygen + morphine. - Answer on NBME exam for how a patient using a self-administering pump develops morphine overdose = "morphine is converted into active metabolites that accumulate." - Otherwise morphine is odd drug for me to put in this table; didn't know where else to put it though.

- 44M + Hx of epistaxis since adolescence + they show you pic of a tongue with a red dot on it + dyspnea + high ejection fraction (75-80%); Dx? → answer on NBME = **pulmonary arteriovenous malformation** → presentation is hereditary hemorrhagic telangiectasia (Osler-Weber-Rendu) → USMLE will always show you a red dot on the tongue or fingernail → Q may also mention fatigue (GI bleeding) → autosomal dominant.
- Child + idiopathic arrhythmia disorder + seizure-like episode; Dx? → Adam-Stokes attack → asked on the peds CMS/NBME form even if you find this menacing, low-yield, or inconvenient → not a true seizure if EEG performed → arrhythmia causes hypoxia of brainstem → seizure-like fit ensues.
- Postmenopausal woman + stress factor + bulging of LV on echo + hypercontractile LV base; Dx? → Takotsubo cardiomyopathy → "ballooning of LV" → once again, weird Dx but USMLE likes it.
- Atherosclerosis; where does the process start? → USMLE answer = endothelial cell, not adipocyte.

- 3F + rumbling murmur auscultated in the neck that goes away **when child is supine and the neck rotated**; Dx? → NBME answer = venous hum → student says wtf? → call it low-yield all you want but it's on the pediatrics CMS/NBME → benign peds murmur that will go away as child grows.
- Congenital heart block; Dx in the mom? → SLE → 1-5% of SLE mothers will have kid with CHB.
- Neonate with supravalvular aortic stenosis; Dx? → kid has William syndrome (chromosome 7, AD; elfin-like facies; hypercalcemia due to increased vitamin D sensitivity; well-developed verbal skills).
- Left ventricular hypertrophy; USMLE asks arrow Q → answer = transcription factor c-Jun activity is increased; beta-myosin heavy chain gene expression is increased; endothelin is increased.
- 65M + 2-3-day Hx of severe chest pain + dyspnea + visible pulsation above manubrium + tracheal deviation + murmur in 2nd intercostal space on the right; Dx? → USMLE answer = aortic aneurysm.
- Dysphagia and/or hoarseness caused by dilated cardiac structure; which structure is dilated? → answer = left atrium → the hoarseness is due to recurrent laryngeal nerve impingement by LA (Ortner syndrome).
- Location of SA node? → answer on NBME = "junction of superior vena cava and right atrium."
- Location of AV node? → answer on NBME = "inferior to the opening of the coronary sinus" → it is located on the posteroinferior interatrial septum near the coronary sinus.
- Location of coronary sinus? → answer = "between the opening of the IVC and the tricuspid valve."
- Neonate with truncus arteriosus; Q is which of the following populations of cells was most likely absent during cardiac development? → USMLE answer = "ectodermal neural crest" cells.
- Fetal alcohol syndrome + heart/lung fistula; mechanism? → answer = "failure of migration of neural crest cells."
- Most common cause of death due to fall or MVA? → traumatic rupture of the aorta (thoracic).
- Where does rupture of the aorta occur? → where the ligamentum arteriosum wraps around the top of the descending arch → ligament is taut but arch is more mobile → leads to shearing.
- What will the NBME/USMLE Q say for traumatic rupture → MVA or fall followed by "widening of the mediastinum on CXR."
- 32M + MVA + widening of mediastinum on CXR; next best step in Dx? → aortic arteriography (aka aortography). New 2CK form also has "CT scan of chest" straight up as answer (i.e., CT angiography). Both show up as answers on different forms, but they won't force you to choose between the two.

- Tx for traumatic rupture? → if ascending arch: labetalol + surgery; if descending arch: labetalol only.
- Traumatic rupture + low BP; next best step? → labetalol (decreases shearing forces, even with low BP it's the answer on the NBME).
- BP different between the arms; Dx? → aortic dissection or subclavian steal syndrome.
- Most likely cause of dissection? → HTN, but connective tissue disorders (e.g., Marfan, Ehlers-Danlos, MYH11 mutations) also important.
- Tx for dissection? → if ascending aortic arch, answer = labetalol + surgery; if descending arch, answer = labetalol only (HY, since everyone chooses surgery) → do **not** choose sodium nitroprusside here.
- Patient with dissection has low BP; next best step in pharm Mx? → labetalol (yes, even with low BP need to decrease shearing forces).
- High TGAs + high chylomicrons on lab report; Dx + mechanism? → familial hyperchylomicronemia; answer on USMLE = "deficiency of apolipoprotein C-II" or "deficiency of lipoprotein lipase."
- Normal TGAs + high LDL on lab report; Dx + mechanism? → familial hypercholesterolemia; answer on USMLE = "deficiency of LDL receptor." If total cholesterol is ~3-500s mg/dL, USMLE wants "**deficiency**"; if total is ~700-1000 mg/dL, the answer = "**absence** of functional LDL receptors on hepatocytes."
- High TGAs + normal LDL on lab report; Dx + mechanism? → familial hypertriglyceridemia; answer on USMLE = "increased hepatic production of VLDL."
- "Creamy plasma" on laboratory inspection. Dx? → familial hyperchylomicronemia.
- Confused old man + temp of 96F + CO high + PCWP low + TPR low; Dx? → septic shock → dementia increases risk of aspiration pneumonia due to diminished gag reflex (important cause of sepsis) → by all means the vignette might say urinary retention in BPH, or tell you there's a catheter → also important causes of sepsis here.
- Patient with infective endocarditis + now has limb weakness or sensory findings; Dx + Tx? → septic embolus → answer = give immediate IV antibiotics.
- Patient with alternating tachycardia + bradycardia; Dx? → sick sinus syndrome → caused by damage to SA node (i.e., due to coronary artery or valvular disease, or autoimmunity, e.g., sarcoidosis) → Tx with atrial pacemaker.
- Diffuse ST-segment elevations on ECG; Dx? → pericarditis.

- If infective, most likely etiology of pericarditis? (answers are bacteria, parasitic, fungal, etc.); answer = viral → viruses can cause serous pericarditis.
- 22M after night of heavy partying + central chest pain worse when leaning back + better when leaning forward; Dx? → serous pericarditis caused by cocaine.
- 68F diabetic + high K + high BUN + high Cr + friction rub in central chest; Dx + Tx? → uremic pericarditis → answer on USMLE = immediate hemodialysis.
- 72M + had STEMI two days ago + now has central friction rub; Dx? → post-MI fibrinous pericarditis.
- 72M + had STEMI 2-6 weeks ago + now has central friction rub; Dx? → Dressler syndrome (autoantibodies causing fibrinous pericarditis).
- 34F + ulnar deviation + MCP/PIP pain + heart problem; Dx? → pericarditis → autoimmune diseases like RA and SLE cause serous pericarditis.
- 34F + anti-Scl70 (topoisomerase I) Abs + heart issue; Dx? → pericardial fibrosis due to systemic sclerosis.
- Kid + heart tumor = cardiac rhabdomyoma until proven otherwise (usually tuberous sclerosis).
- Adult + heart tumor = cardiac myxoma until proven otherwise (ball-in-valve tumor in the left atrium → causes murmur that abates when patient is positioned in an unusual way, e.g., on his right side while leaning diagonally).
- 2-year-old boy has cardiac myxoma (correct, not rhabdomyoma) + perioral melanosis (sophisticated way of saying hyperpigmentation around the mouth/lips) + hyperthyroidism; Dx? → answer = **Carney complex** → **this is asked on the USMLE** → classically triad of cardiac myxoma + perioral melanosis + endocrine hypersecretion (classically bilateral pigmented zona fasciculata hyperplasia resulting Cushing syndrome, but can be hyperthyroidism or growth hormone).
- What does S1 heart sound mean? → closure of atrioventricular valves (mitral + tricuspid), signifying onset of systole.
- What does S2 heart sound mean? → closure of semilunar valves (aortic + pulmonic), signifying onset of diastole.
- What does S3 heart sound mean? → early-diastolic reverberation caused by dilated left ventricle → often synonymous with dilated cardiomyopathy on the USMLE; yes, it can be seen sometimes

incidentally in pregnancy and young athletes (due to increased preload), but almost always it just means dilated cardiomyopathy.

- What does S4 heart sound mean? → late-diastolic reverberation caused by stiff left ventricle → caused by increased afterload on the LV, either due to systemic HTN or aortic stenosis (AS) or hypertrophic obstructive cardiomyopathy (HOCM) → I have seen 2CK-level Qs where an S4 shows up implying **right** ventricular hypertrophy (rare but possible) → for instance patient with mitral stenosis + S4. An S4 on the USMLE is often seen in hypertrophic cardiomyopathy (HCM), which just means a stiff LV; don't confuse HCM with HOCM; HOCM is an AD mutation in the beta-myosin heavy-chain gene causing asymmetric septal hypertrophy; HCM is merely a hypertrophied LV due to increased LV afterload (i.e., due to systemic HTN, AS, or HOCM); in turn HOCM can cause HCM, but they're not the same thing; if HOCM causes HCM, then USMLE loves to give S4 heart sound for that.
- What is a parasternal heave? → a parasternal heave means the heartbeat can be felt (or sometimes seen) along the left sternal border, usually due to RVH (since the RV is most anterior) → RVH can be seen in ventricular septal defect (VSD), so parasternal heave can be seen in VSD.
- What is a palpable thrill? → a palpable thrill is merely a palpable murmur; it carries no additional diagnostic significance; a thrill is seen in grades 4-6 of the heart sounds.
- What are the 6 grades of heart sounds? (not asked on USMLE, but just for your own knowledge with respect to this document) → 1: very faint; not heard in all positions ("cardiologist only"); 2: faint; heard in all positions; 3: loud, with no thrill; 4: loud with palpable thrill; 5: loud with palpable thrill + can be heard with stethoscope partially off chest; 6: loud with palpable thrill + can be heard with the stethoscope completely off the chest.
- Which murmurs are holosystolic (aka pansystolic)? → mitral regurgitation (mitral insufficiency; MR) + tricuspid regurgitation (tricuspid insufficiency; TR); ventricular septal defect (VSD).
- Which murmurs are mid-systolic (crescendo-decrescendo systolic)? → aortic stenosis (AS) + hypertrophic obstructive cardiomyopathy (HOCM) + pulmonic stenosis (PS).
- Which murmur has a diastolic opening snap? → mitral stenosis (MS) → has diastolic opening snap, followed by a mid-late decrescendo diastolic murmur.
- Which murmur has a mid-systolic click? → mitral valve prolapse (MVP).
- Which murmur can also be described as a late-peaking systolic murmur with an ejection click? → AS.

- Which murmur is continuous machinery-like? → patent ductus arteriosus (PDA); can also be AV fistulae.
- Which murmur is pansystolic-pandiastolic? → PDA (same as continuous machinery-like).
- Which murmur is to-and-fro? → PDA; same as continuous or pansystolic-pandiastolic.
- Which murmur is fixed splitting of S2? → atrial septal defect (ASD).
- Which murmurs are holodiastolic (pandiastolic)? → aortic regurgitation (aortic insufficiency; AR) + pulmonic regurgitation (pulmonic insufficiency; PR).
- Which murmur is pandiastolic and loudest in early-diastole? → classically AR (decrescendo holodiastolic murmur).
- Young child + hypocalcemia + harsh systolic murmur at left sternal border; Dx? → DiGeorge syndrome associated with tetralogy of Fallot → on the USMLE, you should essentially think of ToF and DiGeorge syndrome as interchangeable → you can by all means get other heart defects in DiGeorge, e.g., truncus arteriosus, but I can't emphasize enough that ToF is almost always seen in DiGeorge on USMLE.
- How do the heart murmurs change with volume in the heart (i.e., do they get louder/softer with more/less volume)?→ all will get worse / more prominent with more volume in the heart, **however MVP and HOCM are the odd ones out;** they'll get worse with less volume in the heart. No change = softer on USMLE. So for example, AS will be no change or softer with Valsalva. Valsalva decreases volume in the heart.
- 8F + sickle cell + fever + HR of 120 + normal BP + 2/6 mid-systolic murmur at upper right sternal border; Dx? → transient, functional high-flow murmur secondary to tachycardia → called functional because murmur will subside once HR returns to baseline.
- 13F + Hb of 7 g/dL + HR of 120 + normal BP + 2/6 mid-systolic murmur at upper right sternal border; Dx? → once again, transient, functional flow murmur → I point this out because students often erroneously think there's some heart abnormality when they see this type of murmur.
- Aortic stenosis (AS) – what will you auscultate? → mid-systolic (crescendo-decrescendo systolic) murmur classically at 2nd intercostal space, right sternal border, with radiation to the carotids; however will also show up as "late-peaking systolic murmur with an ejection click." → don't confuse the latter with "mid-systolic click," which is mitral valve prolapse (MVP).

- Who gets AS? → classically bicuspid aortic valve → can be familial autosomal dominant; also seen in Turner syndrome (45XO) → leads to early calcification of valve in the 40s onward; however a young patient without significant calcification can easily have AS. USMLE gives Qs of children with bicuspid aortic valve and aortic stenosis and students are somehow surprised by this ("I thought it had to be adults though.")
- What about if the patient doesn't have bicuspid valve? → AS can still occur in the general population with normal senile calcification seen typically age 70s-80s onward (i.e., incidental 1/6 or 2/6 mid-systolic murmur in otherwise healthy elderly patient).
- If patient is diagnosed with bicuspid aortic valve, next best step in Mx? → annual transthoracic echos → if valve cross-sectional area falls below 1.0 cm^2 then do aortic valve replacement; there's a surgery NBME Q where they say cross-sectional area is 0.8 cm^2 and the answer is straight-up "aortic valve replacement."
- How does AS classically present Sx-wise? → SAD → Syncope, Angina, Dyspnea.
- AS causes what kind of LVH? → **concentric hypertrophy due to pressure overload** → can also cause hypertrophic cardiomyopathy with an S4 heart sound (stiff LV). This is in contrast to aortic regurgitation (aortic insufficiency), which causes eccentric hypertrophy due to volume overload.
- What kind of pulse is seen in AS? → slow-rising pulse ("pulsus parvus et tardus"). Don't confuse this with AR, which causes bounding pulses with head-bobbing (Q will often say for AR: "pulse has brisk upstroke with precipitous downstroke.").
- Any weird factoid about AS? → Heyde syndrome is the combo of AS + angiodysplasia (painless rectal bleeding in elderly due to superficial tortuous vessels on the bowel wall) → shows up on NBME.
- What does HOCM sound like? → same as AS (mid-systolic murmur, aka crescendo-decrescendo systolic murmur).
- What causes HOCM? → autosomal dominant mutation in beta-myosin heavy-chain gene.
- What's the structural change in the heart with HOCM? → asymmetric septal hypertrophy that causes the **anterior mitral valve leaflet** to block off the LV outflow tract under states of lesser preload → student says, "if the LV outflow tract is blocked off (i.e., where the aortic valve is), why is it the mitral valve leaflet that blocks it off then?" Yeah, I know, it's weird. But the asymmetric septal hypertrophy causes this to happen.

- What's the cause of death in HOCM? → ventricular fibrillation due to acute left-heart strain → the "sudden death in young athlete" is **not** due to an MI → i.e., the patient has **clean** coronary arteries → do **not** select coronary artery occlusion as the answer.
- What about if the vignette is sudden death in middle-aged patient with heart disease? → answer = ischemic heart disease (MI), not HOCM.
- 18M athlete + 2/6 mid-systolic murmur at right sternal border 2nd intercostal space + there is paradoxical splitting of S2 + there is no change in the murmur with Valsalva; Dx? → **"bicuspid aortic valve" (AS), not HOCM** → students say "oh em gee young athlete! HOCM!" → the USMLE will slam you on this and wants you to know that the key way to distinguish between AS and HOCM murmurs is that HOCM gets worse with lower volume in the heart; AS will soften or there will be no change. Don't just automatically jump on HOCM because it's a young athlete.
- How to Tx HOCM → can give propranolol to keep HR from getting too fast (the slower the HR, the more time the heart spends in diastole → more diastolic filling → greater preload → less occlusion of LV outflow tract) → should be noted tangentially that although beta-blockers increase preload, they decrease chronotropy + inotropy so the net effect is still decreased myocardial oxygen demand.
- Can you explain "splitting of S2"? What does that even mean? → the aortic valve normally shuts (A2) just before the pulmonic valve (P2), so A2 will occur slightly before P2 → when we talk about changes in splitting of the S2 heart sound (i.e., wide splitting, paradoxical splitting, etc.), if pressure in a ventricle is greater, the sound will occur later / is protracted. So if RV pressure becomes greater for whatever reason → P2 occurs later → wider splitting. So pulmonary artery hypertension = wide-splitting. If LV pressure becomes greater → A2 occurs later, and can even occur after P2 → paradoxical splitting. So LVH = paradoxical splitting. When R or L ventricular pressure exceeds the pulmonic arterial and aortic pressure, respectively, the valves open. Then the ventricle will lose pressure as blood ejects, followed by isovolumetric relaxation marking the onset of diastole, and the pressure within the ventricle falls below the pressure distal to the valve → valve shuts. Normally splitting oscillates with the respiratory cycle. Inhalation causes P2 to occur later → decrease in intrathoracic pressure → increased venous return to right atrium → more blood in right ventricle → more preload → more pressure → time it takes for RV pressure to fall below pulmonic arterial pressure is greater → P2 will occur slightly later with inhalation. With exhalation it's the opposite. P2

occurs slightly sooner because increased intrathoracic pressure will attenuate venous return → less preload in RV → less pressure in RV → time it takes for RV pressure to fall below pulmonic arterial pressure is less → pulmonic valve closes slightly sooner → distance between A2 and P2 is less.

- What is fixed splitting of S2? → Super HY for atrial septal defect (ASD) → sometimes can be written as "wide, fixed splitting of S2" → it's not the "wide" that matters; you need to remember **fixed splitting.** Wide splitting just means RVH, which can occur due to chronic volume overload in the setting of some ASDs.
- What does "splitting of S1 mean"? → highly unlikely to show up on the USMLE, don't worry, but for the sake of some people who'd ask, it's usually seen in right bundle branch block (BBB), which causes delayed closure of the tricuspid valve.
- Maneuvers that decrease blood in the heart → Valsalva; standing up from seated position; sitting up from supine position; administration of nitrates → any of these will cause MVP + HOCM to get worse; all other murmurs will soften or not change.
- Maneuvers that increase blood in the heart → Lying down; leg raise while supine; squatting; hand-grip.
- How does Valsalva decrease blood in the heart? → attempted exhalation against a closed glottis → robust increase in intrathoracic pressure → decreased venous return → decreased cardiac preload.
- How do nitrates decrease blood in the heart? → Increased venodilation + venous pooling → decreased venous return to the right heart → decreased cardiac preload.
- How does hand-grip increase blood in the heart? → hand-grip increases afterload → LV cannot eject blood as readily → greater volume of blood left in the LV; it should be pointed out, however, that it would be incorrect to say hand-grip increases preload, as this effect is indirect.
- Exceptions regarding handgrip in terms of murmur intensities? → decreases intensity of aortic stenosis (same as HOCM) + increases intensity of mitral stenosis (in an NBME vignette).
- How does respiration relate to left- vs right-sided murmurs? → inspiration makes right-sided murmurs worse; exhalation makes left-sided murmurs worse.
- Why does inspiration make right-sided murmurs worse? → inspiration → decreased intrathoracic pressure → easier for blood to return to the RA → increased venous return → more preload in right heart → worsening of TR, TS, PR, PS.

- Why does inspiration soften left-sided murmurs? → decreased intrathoracic pressure → increased pulmonary vascular compliance → transient decrease in pulmonary venous return to the LA → decreased LA preload; it should be noted that although RA preload increases, this effect does not carry over to the LA because of pulmonary vascular pooling.
- Why does expiration soften right-sided murmurs? → expiration → increased intrathoracic pressure → harder for blood to return to the RA → decreased venous return → less preload in right heart → softening of TR, TS, PR, PS.
- Why does expiration intensify left-sided murmurs? → expiration → increased intrathoracic pressure → decreased pulmonary vascular compliance → transient increase in pulmonary venous return to the LA → increased LA preload; it should be noted that although RA preload decreases, this effect does not carry over to the LA because of pulmonary vascular compression.
- What does VSD sound like? → Holosystolic murmur at the lower left sternal border. Can sometimes be described as having palpable thrill or heave. Q can also mention a diastolic rumble in addition to the holosystolic murmur due to volume reverberations (I've seen it on NBME).
- If you patch/repair a VSD, what will happen to pressure in the LV, RV, and LA? (up or down arrows) → repairing a VSD will cause up LV, down RV, down LA → the down LA always confuses people → repair of VSD means less blood entering RV → less blood going back through the lungs to the LA.
- Who gets AVSD (atrioventricular septal defect)? → Down syndrome (aka endocardial cushion defect).
- What does ASD sound like and why? → as discussed earlier, fixed splitting of S2 → when you've got an ASD, blood is constantly moving L –> R from LA –> RA (pressure is always greater on the left side). So the effects of inhalation/exhalation are minimized in terms of the A2-P2 split bc you'll always have relatively constant LA → RA flow (and resultant steady RA preload) irrespective of inspiration. The sound can also be described as "wide, fixed splitting" bc of increased RV preload → delayed closure of P2 relative to A2 → slight widening, but it's still fixed for the reasons explained above.
- What is the fossa ovalis? → impression in the interatrial septum following closure of the wall during embryological development. Failure of closure leads to patent foramen ovale, which is a type of ASD.
- ASD/VSD in relation to blood pO2? → USMLE loves making you guess whether you have an ASD or VSD based on info they give you about blood pO2. If they say pO2 increases from SVC to RA, you

know the answer is ASD. If they say pO2 increases from RA to RV, you know VSD is the answer. Path of blood through the heart/lungs (IVC/SVC → RA → RV → pulmonary arteries (deoxygenated) → pulmonary veins (oxygenated) → LA → LV → aorta).

- What does MVP sound like? → as mentioned earlier, mid-systolic click, over 4th intercostal space, left mid-clavicular line.
- Who gets MVP? → most common heart murmur; polygenic; usually benign + incidental; can also get in connective tissue disorders (i.e., Marfan syndrome, Ehlers-Danlos).
- Murmurs seen in connective tissue disorders → MVP or AR.
- What does myxomatous degeneration mean? → answer = MVP → connective tissue degeneration.
- What does MR sound like? → holosystolic (pansystolic) murmur over 4th intercostal space, left mid-clavicular line; classically radiates to the axilla but by all means doesn't have to.
- Most common cause of MR? → ischemia; student says "what do you mean?" → atherosclerosis (i.e., due to diabetes, smoking, HTN, familial) → ischemia → structural heart change (LVH + LV dilatation) + papillary muscle weakening → MVP. This is not the same thing as full-blown papillary muscle rupture following an MI; mere ischemia in the absence of MI can cause MR.
- 68M + T2DM + dilated heart on CXR + S3 heart sound + 2/6 holosystolic murmur over left chest; Dx? → MR due to ischemia from atherosclerosis.
- 68M + T2DM + crushing central chest pain + 3 days later has sudden-onset 4/6 holosystolic murmur over left chest; Dx? → MR from papillary muscle rupture.
- 22M + obvious Marfan syndrome in vignette + stethoscope Q where it starts hovering over aortic valve region; what do you do? → you're listening for either AR or MVP, so if you don't hear anything, just move the stethoscope to the mitral area and you'll hear the mid-systolic click.
- 32M + fleeting/stabbing chest pain along the left-lateral chest wall + has had 20-30 episodes like this in the past + mid-systolic click; Dx? → mitral valve prolapse syndrome → do not need to treat overwhelming majority of the time, even when the patient is symptomatic; on the 2CK NBMEs, they give a symptomatic presentation just like this, and the answer is reassurance/observation, not propranolol.
- What does mitral stenosis sound like? → diastolic opening snap with a decrescendo mid-late diastolic murmur.

- Who gets MS? → 99% due to previous rheumatic fever as a kid.
- What is mechanism for rheumatic fever? → type II hypersensitivity against M-protein of Group A Strep (S. pyogenes) → immune system makes antibodies against Group A Strep M-protein that cross-react with the mitral valve (molecular mimicry).
- How does rheumatic fever present? → **JONES (J♥NES)** → **J**oints (polyarthritis) + **♥** (myocarditis / mitral valve disease) + **N**odules (subcutaneous nodules over bony prominences) + **E**rythema **marginatum** (appears annular [ring-like] and serpiginous [snake-like]; important vocab words actually for medicine) + **S**ydenham chorea (antibody-mediated destruction of corpus striatum of basal ganglia) → my biggest advice here is to remember "marginatum" because it's specifically seen in RF; don't be that person going around saying "RF has……..erythema multiforme…..?" The latter is usually seen as one of the types of drug rashes.
- 12F + red tongue + salmon body rash; Dx and Tx? → scarlet fever caused by Group A Strep. Must give penicillin to prevent RF.
- RF caused by cutaneous Group A Strep? → No. Cutaneous infections (i.e., impetigo, erysipelas, cellulitis) can cause post-streptococcal glomerulonephritis (PSGN), but not RF. RF is caused by pharyngitis / scarlet fever (pharyngitis + body rash).
- 40M + Hx of rheumatic fever as a child; what murmur does he most likely have now? → MS.
- 40M + Hx of rheumatic fever as a child + rumbling diastolic murmur + S4 heart sound; Dx? → MS with a right-ventricular S4 (unusual to be right-sided, but this is on the 2CK NBME).
- 40M + Hx of rheumatic fever as a child + rumbling diastolic murmur; which of the following is most likely part of his Hx as a child? → **mitral regurgitation.** → this is exceedingly HY → **RF causes MR acutely in the child but will cause MS later in life as the valve scars over.**
- 12M + fever + sore throat + painful joints + annular skin rash + heart murmur; most likely murmur that he has? → MR, not MS.
- 12M + fever + sore throat + painful joints + annular skin rash + 2/6 holosystolic murmur over left chest; as an adult, what might we expect in this patient? → answer = diastolic rumbling murmur with opening snap (MS); even though right now he has MR, later in life it will become MS.

- 33F + pregnant at 20 weeks + new-onset dyspnea + crackles in both lung fields + diastolic rumbling murmur; Dx? → mitral stenosis presenting symptomatically now that plasma volume has increased ~50% in pregnancy.
- 33F + pregnant at 38 weeks + prominent bilateral ankle pitting edema + dyspnea; Dx? → peripartum dilated cardiomyopathy (antibody-mediated; idiopathic).
- 33F + pregnant at 32 weeks + mild ankle edema; Dx? → normal edema seen in pregnancy.
- 33F + peripartum dilated cardiomyopathy; next best step in Mx? → transthoracic echo (TTE) to check for ejection fraction (guides management based on severity).
- Does peripartum dilated cardiomyopathy come back in subsequent pregnancies? → Yes, and it gets worse.
- 33F + peripartum dilated cardiomyopathy; best way to predict prognosis if she goes on to have a future pregnancy? → TTE (ejection fraction predicts prognosis for future pregnancy).
- If 99% of MS is due to Hx of rheumatic fever, what else can cause it? → Libman-Sacks endocarditis in SLE is associated with MS. 60% of those with LS endocarditis have anti-phospholipid antibodies (lupus anticoagulant).
- 32F + SLE + diastolic rumbling murmur; most likely characteristic of valvular lesion? → answer = “verrucous vegetations on both sides of the mitral valve” for LS endocarditis.
- 28M IV drug user + 2/6 holosystolic murmur at left sternal border that increases with inspiration + fever; most likely characteristic of valvular lesion? → “large, friable, floppy vegetation” → bacterial endocarditis (tricuspid regurg in this case bc IV drug user).
- New-onset murmur + fever; Dx? → infective endocarditis (IE) until proven otherwise. Unlike rheumatic fever, this is actual bacterial colonization of a heart valve + production of vegetation; RF is a mere antibody response.
- What is acute endocarditis? → no Hx of valve abnormality; caused almost always by **S. aureus**; classically seen in IV drug users); classic new-onset murmur + fever presentation.
- What is subacute endocarditis? → Hx of valve abnormality, i.e., congenital defect, Hx of RF; classically occurs following dental procedures; **S. viridans** (same thing as S. sanguinis or S. mutans) is classic cause; USMLE wants you to know S. viridan’s production of limit dextrins (carbohydrates) enables binding to mitral + aortic valve.

- What is **HACEK**? → Gram (-) organisms that can cause endocarditis – **H**emophilus species, **A**ggregatibacter actinomycetemcomitans, **C**ardiobacterium hominis, **E**ikenella corrodens, and **K**ingella kingae – the yield on the USMLE is extremely low so you do not need to memorize these, but I mention them because students occasionally ask about HACEK + the USMLE likes you to know for some magical reason that Eikenella corrodens is associated with human bites, grows white, and has a bleach-like odor.
- For IE, blood cultures before Abx, or Abx before blood cultures? → Always blood cultures (draw three tubes of blood) before giving Abx.
- Empiric Tx for endocarditis? → Gentamicin (aminoglycoside) PLUS either vancomycin or ampicillin/sulbactam. Add rifampin if patient has prosthetic valve.
- If culture comes back positive for MSSA (not MRSA)? → switch to six weeks nafcillin (highly simplified, but the bottom line is if MSSA is confirmed, USMLE answer is you switch to the beta-lactam).
- Why switch to beta-lactam if MSSA? Why not just stay on vanc? → Beta-lactams are way more efficacious than vanc → vanc is actually not a very good drug, but if the organism causing IE is MRSA, it's first-line.
- Who gets pulmonic stenosis and what does it sound like? → sounds like aortic stenosis (midsystolic murmur) but increases in intensity with inspiration because it's right-sided; classically seen as part of tetralogy of Fallot in DiGeorge syndrome; also seen classically in Noonan syndrome (USMLE will not ask you about Noonan syndrome).
- Who gets pulmonic regurg and what does it sound like? → sounds like aortic regurg (holodiastolic) but increases with inspiration; rare, but can be seen in endocarditis in IV drug users.
- Who gets tricuspid regurg and what does it sound like? → same as mitral regurg (holosystolic murmur) but gets louder with inspiration; highest yield cause on USMLE is pulmonary hypertension / cor pulmonale (these don't cause pulmonic regurg, so sounds a bit weird). Can also be seen in IV drug user endocarditis and carcinoid syndrome.
- Who gets tricuspid stenosis and what does it sound like? → sounds like mitral stenosis presumably (diastolic rumbling murmur, with or without opening snap; but would increase in intensity with inspiration in theory); very rare; I've never seen this in any USMLE question.

- How does isolated left heart failure present? → fluid in the lungs (pulmonary edema) +/- pleural effusion; orthopnea, paroxysmal nocturnal dyspnea (PND); depending on the etiology of the heart failure, the structure of the heart will take on different characteristics, but the important point about LH failure is **fluid in the lungs** → **also really important you know that pulmonary capillary wedge pressure (PCWP) is increased in any LH pathology** (even if the pressure is within the acceptable range prior to full-blown LH decompensation, the PCWP is still increased *relative to the patient's original baseline* in LH pathology).
- What is PCWP? → **equal to left atrial pressure**; if you stick a catheter through the venous circulation all the way back to the right heart, and then into the pulmonary circulation, and then into a distal pulmonary capillary such that it can't go any farther, the pressure reverberations are said to best reflect those of the left atrium. The USMLE is obsessed with PCWP; you need to know it is increased not just in cardiogenic shock, but also in LH pathology as I've stated above.
- What is orthopnea? → reflective of LH decompensation → requirement to sleep on multiple pillows when supine to prevent fluid buildup in lungs; when supine, there's greater venous return → greater preload → worsening of dyspnea because the heart cannot handle the volume (i.e., decompensates) because greater O2 demand by the LV with greater preload.
- What is paroxysmal nocturnal dyspnea (PND)? → like orthopnea, reflective of LH decompensation → severe dyspnea that occurs while sleeping due to redistribution of fluid into the lungs; unlike orthopnea, does not immediately subside when sitting up.
- How does isolated right heart failure present? → right ventricular hypertrophy (unless tricuspid pathology), jugular venous distension (JVD), peripheral edema, hepatosplenomegaly (late).
- What's the most common cause of right heart failure? → left heart failure.
- What is congestive heart failure (CHF)? → right heart failure + left heart failure.
- What causes left heart failure? → systemic HTN, ischemia (atherosclerosis), valvular abnormalities.
- Since left heart failure is the most common cause of right heart failure, what usually causes isolated right heart failure? → lung pathology → when you have lung pathology causing RH failure, that's called cor pulmonale.
- "Wait, can you explain cor pulmonale a little more. I've heard that a lot but don't really understand it." → when you have a lung condition like COPD, cystic fibrosis, chronic asthma, etc., that leads to RH

failure, we call that cor pulmonale. Probably the most important piece of info regarding this condition is that **PCWP is normal,** which tells you the cause of the RH failure cannot be from LH origin. For instance, if you have a guy with COPD who also has heart disease, if his PCWP is elevated, then we cannot conclude that his right heart failure is a result of the lung disease in isolation because increased PCWP can lead to RH failure.

- 45M + 70-pack-year Hx of smoking + JVD + peripheral edema; Dx? → cor pulmonale → signs of RH failure in someone who has lung disease.
- 45M + 70-pack-year Hx of smoking + systemic HTN + JVD + peripheral edema + has crackles in lungs + dilated heart on CXR; Dx? → congestive heart failure; patient has combo of left and right heart failure signs.
- 25M + cystic fibrosis + JVD + peripheral edema + crackles in lungs; Dx? → cor pulmonale → crackles due to CF, not LH failure.
- If USMLE asks you for the mechanism of cor pulmonale, what's the answer? → **pulmonary hypertension** → hypoxic vasoconstriction (e.g., in chronic bronchitis, CF) or obliteration of pulmonary parenchyma (emphysema) will cause a backup of blood and pulmonary HTN → increased afterload on RV → starts the process of cor pulmonale (mere pulmonary HTN is not cor pulmonale; there must be RH failure). Can also be due to fibrosis (e.g., CREST or radiation) + loss of lung parenchyma (emphysema).
- How will USMLE describe pulmonary HTN? → Increased pulmonary vascular markings; loud P2; tricuspid regurg. Loud P2 means valve slams shut due to high pressure distal to it. Highest yield cause of tricuspid regurg on USMLE is pulmonary hypertension / cor pulmonale..
- 28F + non-smoker + dyspnea + JVD + increased pulmonary vascular markings; Dx? → primary pulmonary hypertension.
- What is primary pulmonary HTN? → mutation in *BMPR2* gene leading to narrowing of pulmonary vessels + RH failure.
- Tx for pulmonary HTN → most patients will respond to dihydropyridine CCBs; if fail, can use agents like **bosentan** (endothelin-1 receptor antagonist), prostaglandins (i.e., epoprostenol), or sildenafil (yes, Viagra).

- 28F + non-smoker + dyspnea + JVD + increased pulmonary vascular markings; which of the following might describe her condition? → USMLE answer = increased endothelin-1 expression (which makes sense as we know bosentan is a Tx).
- What is dilated cardiomyopathy (DCM) + what are the causes? → heart failure with dilatation of the LV cavity + **systolic dysfunction** with decreased ejection fraction; classic causes are systemic HTN and ischemia (coronary atherosclerosis), but may also be due to **ABCD** → **A**lcohol (EtOH directly damages myocardium); **B**eriberi (wet beriberi seen in thiamine [B1] deficiency; this is **not** the same as alcoholic cardiomyopathy; it's coincidental that this also occurs in alcoholics; alcoholics can get DCM and need not be B1 deficient); **C**ocaine, **C**hagas disease (Trypanosoma cruzi), **C**oxsackie B virus; **D**oxorubicin (Adriamycin); DCM can also be caused by pregnancy (as discussed earlier) and hemochromatosis.
- How does DCM present? → enlarged cardiac silhouette on CXR (dilated heart), lateralized apex beat (dilated heart); sometimes S3 heart sound; fluid in the lungs (pulmonary edema) +/- pleural effusion; can present with orthopnea, paroxysmal nocturnal dyspnea (PND).
- Is HOCM the same thing as hypertrophic cardiomyopathy (HCM)? → Once again, no. HOCM is an AD condition (as discussed earlier). HCM is the **diastolic dysfunction** of the LV that ensues secondary to increased LV afterload (i.e., from systemic HTN, AS, or HOCM) → the USMLE will often give you an S4 heart sound for HCM.
- What about restrictive cardiomyopathy (RCM)? → **diastolic dysfunction** with failure of the heart to expand; causes are fibrosis, amyloidosis, sarcoidosis, scleroderma, prior radiation, etc.
- What are the important arrows for systolic dysfunction? → **Ejection fraction – decreased; LV pressure – increased; LV volume – increased.**
- What are the important arrows for diastolic dysfunction? → **Ejection fraction – normal; LV pressure – increased; LV volume – normal.**
- "Can you explain restrictive cardiomyopathy vs constrictive pericarditis?" → both are characterized by diastolic dysfunction (decreased ability of heart to expand), but in RCM this is due to myocardial stiffness / inelasticity, whereas in CP, the etiology is strictly pericardial, with TB being the most common cause of chronic constrictive pericarditis; CP is associated with calcification on CXR in about a third of patients; calcification is not seen in RCM.

- 22M + stab wound to left chest + JVD + muffled heart sounds + hypotension; Dx? → cardiac tamponade → Beck triad always seen in USMLE Qs = JVD + muffled heart sounds + hypotension; also associated with pulsus paradoxus.
- What is pulsus paradoxus? → drop in BP of >10 mm Hg on inspiration → reflects inability of the heart to fill → seen in cardiac tamponade, severe lung disease (i.e., severe asthma, COPD), and sometimes severe sleep apnea → my observation is students love to focus on miscellaneous causes of PP, but in reality the USMLE only gives a fuck about cardiac tamponade; Qbank might venture down the asthma route once in a blue moon.
- What's the difference between pericardial effusion and cardiac tamponade? → all tamponades are effusions, but not all effusions are tamponades → a tamponade is merely a pericardial effusion that is associated with hemodynamic decompensation (low BP).
- What determines whether a pericardial effusion becomes a tamponade? → acuteness of fluid accumulation → tamponades can be a small volume that accumulates quickly, i.e., from a stab wound or post-MI LV free-wall rupture; whereas we could have, e.g., a slowly accumulating chylous or serosanguinous accumulation in the setting of lymphoma/malignancy that is large volume but does not lead to tamponade.
- USMLE asks where pericardial fluid is secreted into (Step 1) → between the visceral and parietal serous layers → the pericardium = visceral + parietal serous layers + an outer fibrous layer.
- 6M + strong radial pulses + cold lower extremities; Dx? → coarctation of the aorta; coarctation need not only occur in Turner syndrome.
- What is pre-ductal vs post-ductal coarctation? → if the coarctation occurs proximal to the ductus arteriosus insertion on the descending aortic arch, it's called pre-ductal; this will be a very sick/cyanotic baby. In contrast, post-ductal will be the answer for a school-age child or adult who merely has BP/pulses that are greater in the upper extremities.
- Congenital rubella syndrome + cardiovascular defect; Dx? → PDA → continuous machinery-like murmur, pansystolic-pandiastolic, or to-and-fro; can also present with bounding pulses similar to AR.
- How to close PDA? → indomethacin (NSAID).
- How to keep PDA open? → prostaglandin.

- Kid is cyanotic at birth; before surgery, what should be given? → prostaglandin → open PDA can mask and/or mitigate the effects of cyanotic congenital heart disease.
- Bile acid sequestrants; important info? → colestipol, cholestyramine, colesevelam → decrease enterohepatic circulation of bile acids → liver needs to make more → liver pulls cholesterol out of the blood in order to convert it to more bile acids → decreases serum LDL → may slightly increase TGAs → just be aware of their names + mechanism for Step 1 → these drugs do not decrease mortality.
- Ezetimibe; mechanism? → inhibits absorption of cholesterol through the small bowel wall at brush border → does not decrease mortality → just be aware of this drug name + MOA for Step 1.
- Fibrates; important info? → gemfibrozil, fenofibrate → upregulate lipoprotein lipase (LPL) → increases TGA clearance out of blood → **answer on the USMLE is TGAs >300 mg/dL; first-line agent to treat severe hyper-TGAs** → also upregulates PPAR-alpha, which increases HDL synthesis → cause hepatotoxicity (same as statins), myositis with increased creatine kinase (CK; especially when combined with statins), and cholesterol gallstones (due to inhibition of 7-alpha-hydroxylase, which converts cholesterol to bile acids) → do not decrease mortality.
- Why is myositis/rhabdomyolysis more likely when fibrates are combined with statins? → USMLE wants “inhibition of P-450 enzymes” as the answer (fibrates inhibit P-450).
- Statins; mechanism? → competitive, reversible inhibition of HMG-CoA reductase → this secondarily causes upregulation of hepatic LDL receptors → both mechanisms decrease serum LDL → statins decrease morality, not because of their LDL-lowering effect (because other drugs do that too), but because they have an antioxidant effect that transcends the cholesterol-lowering effect → USMLE wants you to know that statins **increase HMG-CoA mRNA synthesis** (compensatory; makes sense, but students get the Q wrong; this is on an NBME); cause hepatotoxicity and myositis/rhabo (the latter especially when combined with fibrates, as said above).
- Orlistat; MOA → pancreatic lipase inhibitor → sometimes used to Tx obesity → can cause fat-soluble vitamin deficiencies + steatorrhea.
- Evolocumab + alirocumab; MOA? → PCSK9 inhibitors → Proprotein convertase subtilisin/kexin type 9 is an enzyme that breaks down LDL receptors → therefore these drugs prevent breakdown of LDL receptors and decrease LDL cholesterol by enabling greater clearance → these are newer agents than

statins and are fair game on the USMLE → New England Journal of Medicine study showed they decrease mortality.

- Digoxin; MOA? → inhibits Na/K-ATPase pump → leads to buildup of Na inside the cardiac myocyte → leads to indirect inhibition of Na/Ca-ATPase pump bc the buildup of intracellular Na disfavors the inward movement of Na via this latter pump → therefore Ca doesn't move out of the cell → increased intracellular Ca → increased inotropy/contractility → also slows HR via Vagal stimulation.
- Yellow wavy vision + drug OD? Dx? → digoxin toxicity.
- How to Tx digoxin toxicity? → normalize potassium + give anti-digoxin Fab fragments + give Mg.
- Electrolyte disturbance causing digoxin toxicity? → hypokalemia (digoxin normally binds to extracellular K+ binding site on the cell, so low K+ means more digoxin binding).
- What are the Ia Na channel blockers? → "the **Q**ueen **P**roclaims **D**iso's pyramid." → **Q**uinidine, **P**rocainamide, **D**isopyramide → you need to know quinidine causes cinchonism (think "quinchonism"), which is tinnitus + headache → **procainamide causes DILE** (drug-induced lupus erythematosus, with anti-histone antibodies; really HY!) → procainamide is the answer on the USMLE for the drug used to Tx Wolf-Parkinson-White syndrome (delta wave on ECG) → disopyramide is ultra-LY and unlikely to show up, but it's mentioned in nearly every resource and completes the mnemonic well → the Ia Na channel blockers increase the risk of torsades de pointes (TdP; a sinusoidal ventricular arrhythmia with high chance of progression to VF and death) → they also increase the action potential duration.
- What are the I**b** Na channel blockers? → "I'd **b**uy **L**iddy's **M**exican **T**acos." → **L**idocaine, **M**exiletine, **T**ocainide → highest yield detail is that they cause "CNS stimulation/depression"; in other words, you'll get a Q where a patient was started on an anti-arrhythmic and gets, e.g., delirium, and then they'll ask you for the drug, which will be, e.g., mexiletine → these agents preferentially act on ischemic tissue → Ibs shorten the action potential duration.
- What are the Ic Na channel blockers? → Flecainide, Encainide, Propafenone → highest yield detail is that flecainide is the first-line anti-arrhythmic to Tx atrial fibrillation in the absence of structural or coronary artery disease → Ics do not change action potential duration → do not use post-MI.

- What are the type II anti-arrhythmics? → beta-blockers → HY mnemonic for agents that antagonize beta-1 only are "**A BEAM** of beta-blockers" → **A**tenolol, **B**isoprolol, **E**smolol, **A**cebutolol, **M**etoprolol → metoprolol used largely as rate control for atrial fibrillation + flutter.
- What are the beta-blockers that also act on alpha-receptors? → carvedilol + labetalol antagonize both alpha- and beta-receptors.
- Which beta-blockers decrease mortality in heart failure? → nebivolol, carvedilol, bisoprolol, metoprolol XR.
- Notable uses for propranolol? → essential tremor (AD/familial, bilateral resting tremor in young adults); migraine prophylaxis (on FM CMS/NBME, is the answer in patient with HTN + migraines); tachycardia in hyperthyroidism (beta-blockade decreases peripheral conversion of T4 to T3); esophageal varices prophylaxis (decreases portal blood flow); akathisia (due to anti-psychotics); HOCM (increase end-diastolic filling → decrease murmur); social phobia; infantile hemangiomas (only if patient with severe complications).
- Notable use of timolol? → topical solution used for glaucoma → decreases aqueous humour production.
- Notable side-effects of beta-blockers? → patients with depression + sexual dysfunction.
- Contraindications to beta-blockers → severe lung disease (asthma w/ Hx of hospitalization or O2 use; COPD); severe or psychotic depression; 2nd or 3rd degree heart block; symptomatic bradycardia; use cautiously in severe diabetes due to masking of hypoglycemic events.
- Tx for beta-blocker overdose? → glucagon.
- Important point about BB in relation to pheochromocytoma? → Never give beta-blocker before alpha-blocker; always give phenoxybenzamine (irreversible alpha-blocker) before beta-blocker in pheo.
- What are the type III anti-arrhythmics? → potassium channel blockers → amiodarone, dronedarone, sotalol, ibutilide, dofetilide → often used in ventricular arrhythmias and in patients being defibrillated unsuccessfully → can be used in atrial fibrillation for rhythm control in those with structural heart disease who cannot take flecainide.

- Points about amiodarone? → causes TdP, pulmonary fibrosis, drug-induced thyroid dysfunction (~40% iodine by weight), corneal deposits, blue-grey skin discoloration → do PFTs, TFTs, LFTs before commencing amiodarone.
- What are the type IV anti-arrhythmics? → non-dihydropyridine calcium channel blockers → verapamil, diltiazem → verapamil is aka non-dihydropyridine CCB → acts selectively on cardiac (nodal) tissue; in contrast, dihydropyridines like nifedipine act on vascular smooth muscle; diltiazem is said to be a mixed agent → verapamil's notable use is for AF rate control in patients who cannot receive beta-blockers → verapamil causes **constipation (really HY)**; it can also cause hyperprolactinemia; diltiazem has occasional utility in the Tx of anal fissure and achalasia.
- What is ivabradine? → inhibits myocardial "funny" Na channels (phase IV of AP) → may be attempted in select patients with stable angina who cannot take beta-blockers → can cause "luminous phenomenon," which is brightness in an area of the visual field.
- What is sacubitril? → neprolysin inhibitor (now you're like, "wtf is neprolysin?") → neprolysin is an enzyme that breaks down ANP and BNP → therefore sacubitril can be used as an antihypertensive to allow the kidney to excrete sodium and water → sacubitril usually used in combination with valsartan as sacubitril/valsartan in the Mx of heart failure (i.e., can be used in place of ACEi or ARB monotherapy during the initial Tx of HF).
- Tx of heart failure? → Start with ACEi, ARB, or sacubitril + ACEi/ARB (beta-blocker first is wrong answer on USMLE) → then make sure patient is euvolemic with furosemide → then add beta-blocker (metoprolol XR, bisoprolol, carvedilol, or nebivolol; these four decrease mortality in HF) → if EF still low, add spironolactone → if EF still low, add **COMBO** of hydralazine + nitrates (combo decreases mortality, especially in African Americans; USMLE Q will mention CHF patient on like 12 meds, and you'll see hydralazine is one of them, which isn't typical, and then they'll ask how the pharm regimen can be modified to decrease morbidity / risk of mortality, and the answer = "add isosorbide ditrate") → if EF still low, add digoxin → if EF still low, use implantable defibrillator.
- Super important point about above sequence? → **Digoxin and furosemide DO NOT decrease mortality** (USMLE is obsessed with that point).
- What is ranolazine? → sodium channel blocker used rarely for angina.

- Mechanism of nitrates? → NO activates guanylyl cyclase → increased cGMP production → increased protein kinase G → increased dephosphorylation of myosin light-chain kinase → vascular smooth muscle relaxation → work on veins >> arteries → venodilatation is main effect → must not use with sildenafil (sildenafil inhibits PDE-5 → prevents breakdown of cGMP → severe hypotension with nitrates).
- What about sodium nitroprusside? → works on both arteries and veins → used in hypertensive emergencies → can cause cyanide toxicity → if patient with HTN emergency gets confusion after administration of SN, the answer = cyanide toxicity; if patient has confusion before administration, answer = hypertensive encephalopathy.
- What is fenoldopam? → dopamine 1 (D1) receptor agonist → dilates both renal afferent and efferent arterioles → used to maintain renal perfusion in hypertensive emergency.
- 66M + back pain + hypercalcemia + renal insufficiency + Q shows you a pic of a white, fibrotic-appearing heart; Dx? → answer = cardiac amyloidosis → seen classically in multiple myeloma (renal insufficiency is often renal amyloidosis).
- Hypovolemic shock arrows → CO down, VR down, TPR up, PCWP down (or normal).
- Cardiogenic shock arrows → CO down, VR down, TPR up, **PCWP up.**
- Septic + anaphylactic shock arrows → CO up, VR up, TPR down, PCWP normal.
- Neurogenic shock + adrenal crisis arrows → CO down, VR down, TPR down, PCWP normal.
- "Can you explain 'autoregulation'?" → Carotid sinus baroreceptors (think "sinus pressure") are stretch-dependent (higher BP = more stretch = more firing of CN IX; **lower BP = less stretch = less firing of IX**) → so if, for instance, there is high BP, we get increased CN IX (glossopharyngeal) afferent firing to solitary nucleus (nucleus solitarius) of the medulla → result is increased CN X parasympathetic efferent firing to atria (M2 receptors) to slow HR + decreased sympathetic outflow.
- "I'm still a little confused though. If patient is hypovolemic, then what happens in terms of autoregulation?" → decreased stretch of carotid sinus baroreceptors → **DECREASED** afferent firing of CN IX → decreased efferent CN X firing → decreased ACh binding to M2 receptors at atria + increased sympathetic outflow → HR increases.
- Mediator that causes cardiac pain? → Adenosine. Produced from ATP breakdown. In the setting of ischemia, adenosine builds up, causing cardiac pain.

- Main regulators of coronary blood flow? → O2, CO2, adenosine, NO. But adenosine is most important. Q will tell you norepinephrine was administered, resulting in reduced coronary artery blood flow. Then very soon after blood flow drastically increases. They ask what's responsible. Answer = adenosine. Presumably NE causes increased oxygen demand, resulting in ATP usage and adenosine production → increased blood flow.
- Main regulators of skeletal muscle blood flow? → "CHALK" → CO2, H+, adenosine, lactate, K+.
- Q asks if coronary blood flow is greater or equal during systole or diastole when person is at rest vs exercising → answer = coronary blood flow is: diastole > systole regardless if patient is at rest or exercising.
- ST-elevations in leads II, III, aVF; which vessel is affected? → answer = posterior descending artery → supplies inferior portion of the heart → sometimes the answer will just be straight-up "right coronary artery" (~70-80% of the time the PDA comes off the RCA).
- ST-elevations in leads V1, V2, V3; which vessel is affected? → left anterior descending artery (LAD; aka anterior interventricular artery) → supplies anterior heart.
- ST-**depressions** in leads V1, V2, V3; which vessel is affected? → posterior descending artery → posterior infarcts can present as reciprocal ST-depression in the anterior leads.
- ST-elevations in V4, V5, V6; which vessel is affected? → left circumflex artery (LCx) → supplies left lateral heart.
- Hypokinesis of the apex of the heart on echo; which vessel is affected? → LAD → supplies apex.
- Wide-complex tachycardia → ventricular tachycardia (VT).
- Narrow-complex tachy → SVT.
- Tx for SVT → vagal/carotid massage ("vagal maneuvers") first; if doesn't work, then adenosine.
- Tx of VT → anti-arrhythmics, e.g., amiodarone.
- Tx of SVT or VT in setting of coma or low BP → direct-current countershock.
- Tx of first-degree heart-block or second-degree Mobitz I (Wenckebach) → observe.
- Tx of second-degree Mobitz II or third-degree heart block → pacemaker.
- First-degree heart-block → PR-interval >200 milliseconds.
- Mobitz I → gradually prolonging PR-interval before a QRS drops.

- Mobitz II → no gradual prolongation of PR-interval; QRS randomly drops.
- Third-degree → HR super slow at 30-40; no relation between p-waves and QRS complexes.
- Infective causes of third-degree → Lyme disease, congenital lupus, diphtheria.
- Mechanism of Goodpasture syndrome? → antibodies against type IV collagen → "2, 3, 4… 2, 3, 4… 2, 3, 4. The Goodpasture is marching in the field, 2, 3, 4!"→ Type 2 hypersensitivity against the alpha-3 chains of type 4 collagen. Presents as male 20s-40s with hemoptysis + hematuria.
- Hematuria + hemoptysis + "head-itis" (mastoiditis, sinusitis, otitis, nasal septal perforation) → Wegener granulomatosis.
- Annoying new name for Wegener → granulomatosis with polyangiitis.
- Dx of Wegener → c-ANCA (anti-PR3; anti-proteinase 3).
- Asthma + eosinophilia → Churg-Strauss.
- Annoying new name for CS → eosinophilic granulomatosis with polyangiitis.
- Dx of CS → p-ANCA (anti-MPO; anti-myeloperoxidase).
- Hematuria in isolation + p-ANCA in serum → microscopic polyangiitis (MP).
- Severe renal disease in Wegener or Goodpasture or MP → rapidly progressive glomerulonephritis (crescentic).
- 44M + hematuria + hemoptysis → Goodpasture syndrome.
- 44M + hematuria + hemopytisis + head-itis → Wegener.
- What is polyarteritis nodosa → medium-vessel vasculitis characterized by immune complex deposition in vascular walls + fibrinoid necrosis (can be described as "segmental, ischemic necrosis").
- Polyarteritis nodosa is associated with what infection? → 30% of patients are HepB positive.
- What do you see on renal artery angiogram in PN → "beads on a string" (similar to fibromuscular dysplasia, although completely unrelated diseases).
- Kid jumps into cold lake; what happens to (arrows) central blood volume, ADH, and ANP levels? → answer = central blood volume increases, ADH decreases, ANP increases → cold means increased alpha-1 agonism in arterioles to constrict distally to retain heat → increased blood volume in larger vessels → increased right atrial filling + stretch → increased ANP release; increased central blood volume also increases baroreceptor activity at carotid sinus → not only leads to increased activity of

the autoregulation CN IX/X loop, but also suppresses ADH release to decrease free water reabsorption in the medullary collecting duct of the kidney. If this part sounds confusing, you can remember the soccer-mom explanation instead, which is just: if you have more volume in your core, your body "thinks" it doesn't want more volume, so ADH goes down to decrease free water reabsorption in the kidneys.

- Medial malleolus ulcer + hyperpigmentation of lower legs; Dx? → chronic venous insufficiency
- Punched-out ulcer on foot + intermittent claudication; Dx? → arterial insufficiency (peripheral vascular disease)
- What causes venous insufficiency? → valvular incompetence (most commonly familial), resulting in venous reflux + insufficiency. Patient need not have varicose veins. Venous insufficiency can present as hyperpigmentation due to hemosiderin deposition and Hx of DVT.
- What causes arterial insufficiency → atherosclerosis (diabetes, followed by smoking, are the two most acceleratory risk factors; hypertension is the most common risk factor)
- How do you Dx venous insufficiency? → **duplex ultrasound** of the calves showing stasis and/or occlusive disease.
- How do you Dx arterial insufficiency? → USMLE always wants **ankle-brachial indices (ABI) first** → After this is done, the answer is **exercise stress test** (to determine exercise tolerance). If not listed, choose **exercise/walking program.** Do not choose arteriography or cilostazol as immediate answers.
- Tx for venous insufficiency → compression stockings
- Tx for varicose veins → compression stockings
- Varicose veins and venous insufficiency same thing? → varicose veins are one of the mere presentations of venous insufficiency, so yes, patients with varicose veins have venous insufficiency.
- 47F has varicose veins + painful palpable cord by the ankle (is the treatment compression stockings or subcutaneous enoxaparin; both are listed) → answer = subcutaneous enoxaparin because this is superficial thrombophlebitis.
- Tx for arterial insufficiency → exercise regimen first, THEN cilostazol (phosphodiesterase 3 inhibitor)
- What must you do before starting the exercise regimen in the Tx of arterial insufficiency → **ECG stress test** to ascertain patient's exercise tolerance.
- What if patient has abnormal baseline ECG (e.g., BBB) → do echo stress test instead.

- What if the patient can't exercise → do dobutamine-echo stress test
- What if the patient gets stable angina after merely walking up a flight of stairs → skip stress test and go straight to myocardial perfusion scan (myocardial scintigraphic assay); this is answer on the NBME.
- Patient has severe ischemia on stress test or myocardial perfusion scan → do coronary angiography → then do coronary artery bypass grafting if three-vessel disease, OR two-vessel disease + diabetic, OR single-vessel disease if it's the left main coronary.
- Patient with CVD is on various medications + has hyperkalemia; why? → ACEi, ARB, and spironolactone all can cause hyperkalemia.
- Patient with CVD is on various medications + hypokalemia; why? → furosemide (Loop diuretic)
- When do we start patients on furosemide? → to fluid unload (dyspnea in heart failure or peripheral edema)
- Patient is started on furosemide + still has fluid overload; what's the next diuretic to use → spironolactone (this is really HY on the USMLE and is on Steps 1 and 2CK NBMEs) → essentially furosemide causes increased K wasting, so we must give a potassium-sparing diuretic to balance the effect (spironolactone).
- What's the MOA of spironolactone → aldosterone receptor antagonist.
- Side-effects of spironolactone → hyperkalemia; gynecomastia.
- When do we give patients spironolactone apart from as a step-up from Loops? → added onto heart failure management after a patient is already on ACEi (or ARB) + beta-blocker. In other words, for heart failure: give ACEi (or ARB) first, then add beta-blocker, then add spironolactone.
- Major side-effects of beta-blockers → depression + sexual dysfunction (avoid in these patients)
- Major side-effect of naproxen → fluid retention (edema) due to increased renal retention of sodium.
- What is naproxen? → NSAID that the USMLE is obsessed with for some reason.
- Why might NSAIDs cause fluid retention / renal retention of sodium? → knocking out COX → decreased prostaglandin synthesis → decreased renal afferent arteriolar dilatation → decreased renal blood flow → PCT of kidney compensates for perceived low blood volume by increasing Na reabsorption → water follows sodium → edema.
- Most common cause of carotid plaques? → HTN → the strong systolic impulse from the heart pounds the carotids --> endothelial damage --> atherosclerosis.

- 55M + BP 150/90 + TIA; next best step in Mx? → carotid duplex USS → the first thing you want to think about is, "does this guy have a carotid plaque that has resulted in a clot embolizing to his brain."
- 80M + good blood pressure (e.g., 110/70) + stroke or TIA; next best step in Mx? → ECG → you want to think, "Does he have atrial fibrillation with a LA mural thrombus that's now embolized to the brain."
- 80M + good blood pressure (e.g., 110/70) + stroke or TIA + ECG shows sinus rhythm with no abnormalities; next best step in Mx? → Holter monitor → when you first see this scenario you're probably like, "Wait, the ECG is normal, so it's not AF?" → No, *it is* likely AF, but AF is often paroxysmal, so in order to detect it in this scenario, **the next best step is a Holter monitor (24-hour wearable ECG).** This means that later in the day when he sits down to have dinner and then pops into AF, the Holter monitor will pick it up.
- What % of people over age 80 have AF? → 8% of people over age 80 have AF, which is why age is a huge risk factor. In other words, if the vignette says the guy is 58, AF is probably less likely just based on shear probability, regardless of hypertensive status." And, once again, knowing that AF is often paroxysmal is really important.
- Age 50s-60s + high BP + TIA/stroke/retinal artery occlusion; next best step in Dx? → answer = carotid duplex ultrasound to look for carotid plaques.
- Age >75 + good BP + TIA/stroke/retinal artery occlusion; answer = ECG to look for AF → if normal, do Holter monitor (24-hour ambulatory ECG monitor) to pick up paroxysmal AF.
- 55M + good BP + carotid bruit heard on auscultation; next best step in Mx? → answer = carotid duplex ultrasound to look for carotid plaques → in this case, if they are obvious and explicit about the suspected etiology of the stroke, TIA, or retinal artery occlusion, then you can just do the carotid duplex ultrasound.
- How to Mx carotid plaques? → First we have to ask whether the patient is symptomatic or asymptomatic. A bruit does not count as symptoms (that's a sign). Symptomatic means stroke, TIA, or retinal artery occlusion. According to recent guidelines: carotid occlusion >70% if symptomatic, or >80% if asymptomatic → answer = do carotid endarterectomy. Below these thresholds → answer =

medical management = triad of 1) ACEi or ARB, 2) statin, 3) anti-platelet therapy (aspirin alone sufficient for USMLE).

- For the actual Tx of the AF, we do rate control before rhythm control (the management is actually heavily involved, but for the USMLE know the following):
    - **Rate control:** beta-blocker first-line (metoprolol). If beta-blocker avoided (i.e., severe or psychotic depression, sexual dysfunction, COPD, Hx of asthma requiring oxygen or hospitalization, 2nd/3rd-degree heart block), verapamil is the next choice. If rate control fails, go to rhythm control.
    - **Rhythm control:** Flecainide (type-Ic Na channel blocker) first-line in those without any structural (i.e., LVH or valvular problems) or coronary artery disease (any symptomatology of CVD or PVD means patient has coronary artery disease). In those who cannot receive flecainide, other anti-arrhythmics like amiodarone, dronedarone, and dofetilide may be used.
- 68F + diabetic + diffuse, dull abdo pain 1-2 hours after meals; Dx? → chronic mesenteric ischemia due to atherosclerosis of SMA or IMA, **not** duodenal ulcer (if they want the latter, they'll say 29M from Indonesia) → essentially stable angina of the bowel.
- 68F + Hx of intermittent claudication + CABG + abdo pain 1-2 hours after eating meals; Dx? → chronic mesenteric ischemia.
- 78M + Hx of AF + acute-onset severe abdo pain "out of proportion to physical exam"; Dx? → acute mesenteric ischemia due to embolus (LA mural thrombus that's launched off to SMA or IMA).
- 66F + has episode of ventricular fibrillation + defibrillated + now has severe abdo pain; Dx? → acute mesenteric ischemia → defibrillation likely launched off LA mural thrombus the patient had (even if no overt Afib).
- 68F + diabetic + Hx of diffuse, dull abdo pain 1-2 hours after meals + now has 2-day Hx of severe abdo pain out of proportion to physical exam; Dx? → acute on chronic mesenteric ischemia due to ruptured atherosclerotic plaque (akin to an "MI" of the bowel).
- Dx of acute + chronic mesenteric ischemia? → USMLE answer = mesenteric angiography.

- Tx of acute mesenteric ischemia? → endarterectomy might be able to restore blood flow if caught in time, but on the USMLE, they will say "IV antibiotics are administered; what's the next best step in Mx?" and the answer is just "laparotomy."
- Tx of chronic mesenteric ischemia → endarterectomy to clear vessel.
- 45M + has FHx of DCM + cirrhosis + generalized hyperpigmentation; his heart may show accumulation of what? → answer = Iron. Diagnosis is hereditary hemochromatosis, which can present as "bronze diabetes" → 1) increased glucose levels + 2) hyperpigmentation + 3) other miscellaneous finding like cardiomyopathy, or pseudogout, or infertility.
- 92F dies in her sleep; heart is most likely to show what on biopsy? → lipofuscin → yellow-brown "age pigment" that reflects autodigestion of intracellular lipid residues by lysosomal enzymes.
- Post-MI dyspnea; mechanism for fluid in lungs? → answer = "increased pulmonary capillary pressure" → the wrong answer is "increased permeability of pulmonary capillaries." The former refers to transudate; the latter refers to exudate.
- Left dominant coronary circulation in someone with STEMI in leads II, III, and aVF; what is the pathway of vessels here? → firstly, need to identify this as inferior infarct; then identify that the posterior descending artery (PDA) supplies the inferior portion of the heart; Q gives you "LCA," "LCx" and "PDA" as answers in different order; answer = LCA → LCx → PDA (left-dominant means must start with LCA; we know PDA finishes the supply, so LCx must be in the middle).
- Patient has end-systolic volume of 100 mL + end-diastolic volume is 150 mL; what's the ejection fraction in terms of %? → answer = (EDV – ESV / EDV) = (150 – 100 / 150) = 50/150 = 33.3% → USMLE will ask you to calculate.
- What is normal ejection fraction? → 55-70%.
- When might the Q say the EF is high? → sometimes in high-output cardiac failure due to AV shunts, i.e., in Paget disease of bone, hereditary hemorrhagic telangiectasia (pulmonary AVM), or patient with previous dialysis + shunt.
- 29F + SOB with exertion + S3 heart sound + laterally displaced apex beat + CXR shows cardiomegaly + Kerley B lines + TTE shows EF of 30%; Dx? → answer = CHF → you're supposed to say, "Wait, in a young patient without any specific disorder? And they don't mention pregnancy either?" → but the rest of the presentation is overwhelming, so you need to know this is ultra-classic / HY for CHF.

- NBME Q on CHF asks for (up or down) for LV stroke volume, LA pressure, and TPR → answer = LV stroke volume down; LA pressure up; TPR up.
- 56M + MI + has coronary angioplasty to restore blood flow + now has idioventricular arrhythmia; why? → answer = “generation of reactive oxygen species”; this refers to reperfusion injury.



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